

Computational Modeling of the HyMAX Experiment in Support of a New Aeroelastic Experimental Design

07/22/25

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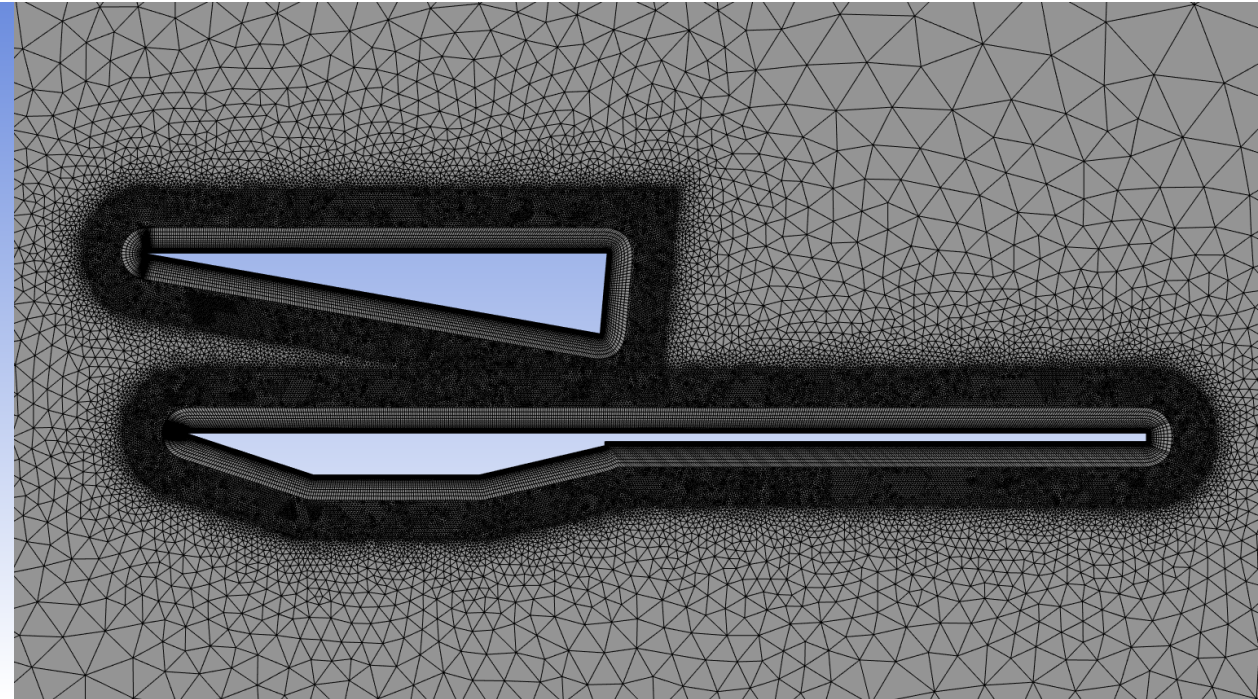
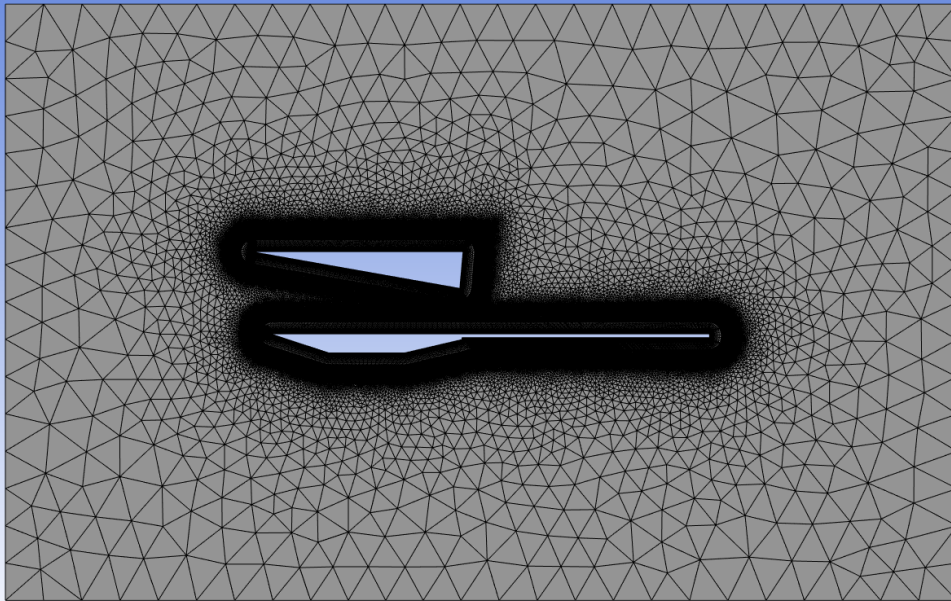
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Background

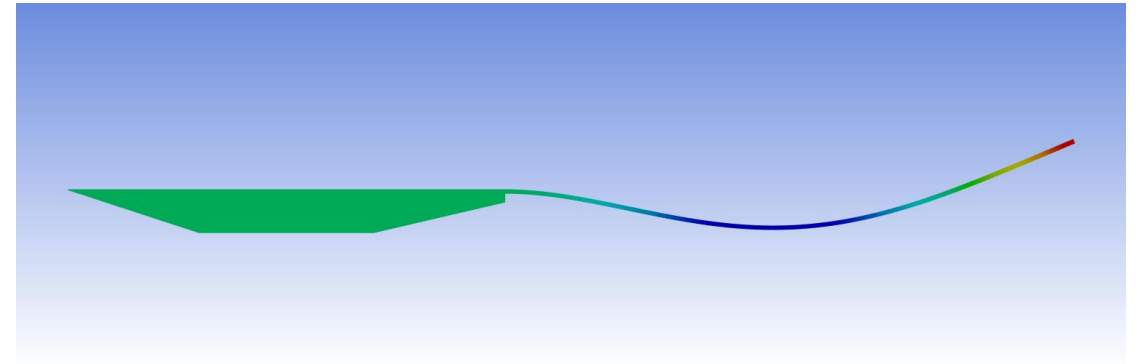
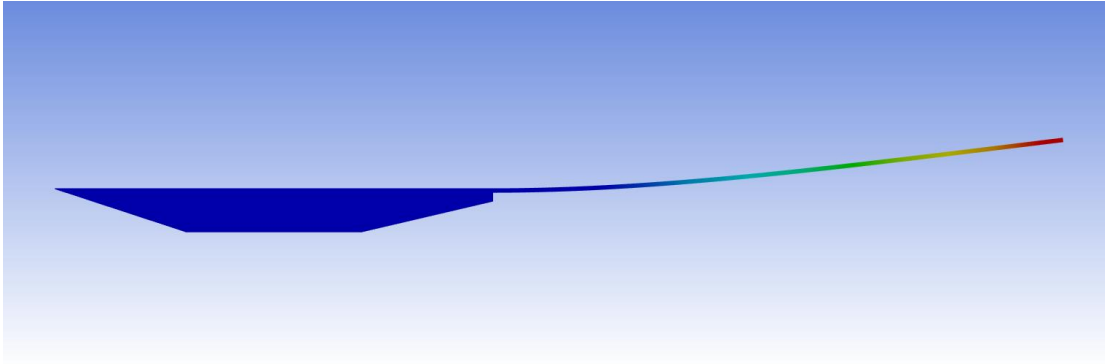
- One of two test cases for the High-Speed Working Group
- A hypersonic cantilevered plate experiment performed by University of New South Wales (UNSW) Canberra at The University of Southern Queensland (TUSQ) Ludwig tube
 - A flexible cantilevered plate attached to a stand
 - $L = 130 \text{ mm}$, $h = 2 \text{ mm}$, $b = 80 \text{ mm}$
 - Optional shock generator above the setup
 - 2 primary cases – 2 degrees and 10 degrees
 - Optional – Oscillating wedge (5 degrees $\pm$ 3 degrees)
 - Test Conditions (freestream values)
 - $M = 5.8$
 - $T = 75 \text{ K}$
 - $P = 755 \text{ Pa}$
 - $Re = 7.1 \text{e}6 / \text{m}$

- Solver used was FUN3D v14.2
 - 2D Mode, Riemann (farfield) BC on the edges of the domain, symmetry BC on the spanwise boundaries, no-slip walls
 - Turbulence Model: SA-neg with Reynolds Stress Model (QCR2000) and Rotation and Compressibility corrections
 - ALDFSS flux construction, Venkatakrishnan limiter
 - Optimized second-order backward differencing in time
 - $\Delta t = 1\text{e-}3$ ($5.76\text{e-}6$ s)
 - Referenced off prior study by Bartels¹
- Utilized built-in aeroelastic modal solver
 - Two step process
 - Obtain a steady state solution with no motion
 - Restart from steady flow field with motion enabled and zero applied structural damping

- Unstructured mesh with inflation layer
 - Coarse mesh in the farfield
 - Two areas of refinement – around the shock generator and test section
 - Inflation layer with $Dz = 3e-6$ m the on the shock generator and test section
 - $y^+ < 1$ along the plate



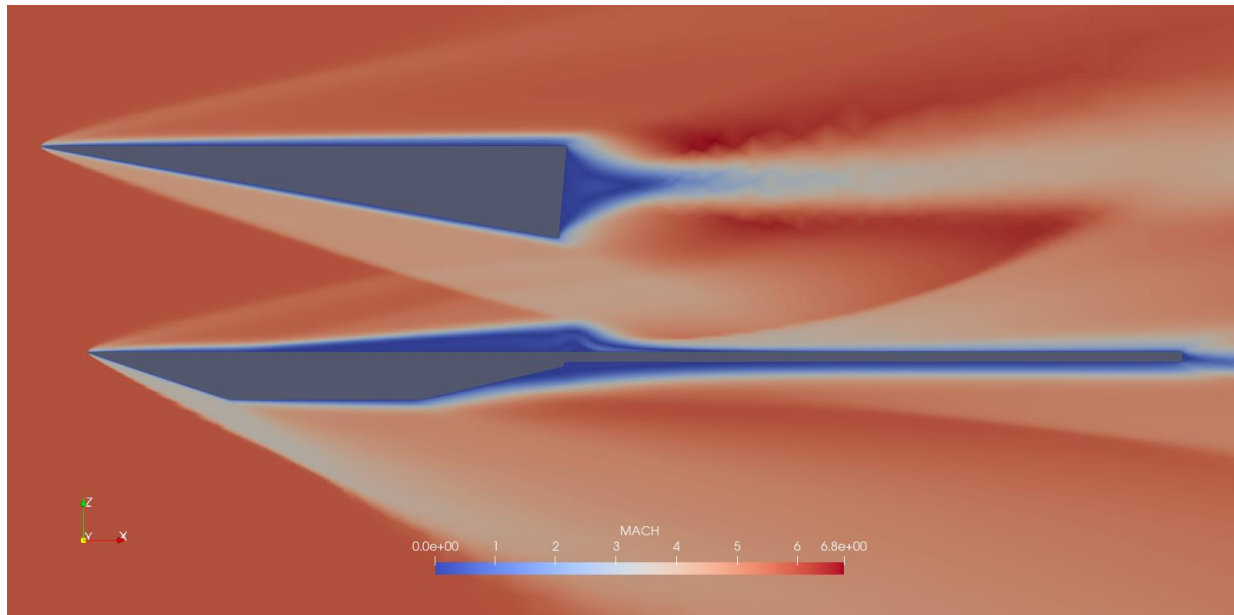
- Only two structural modes used in the simulations



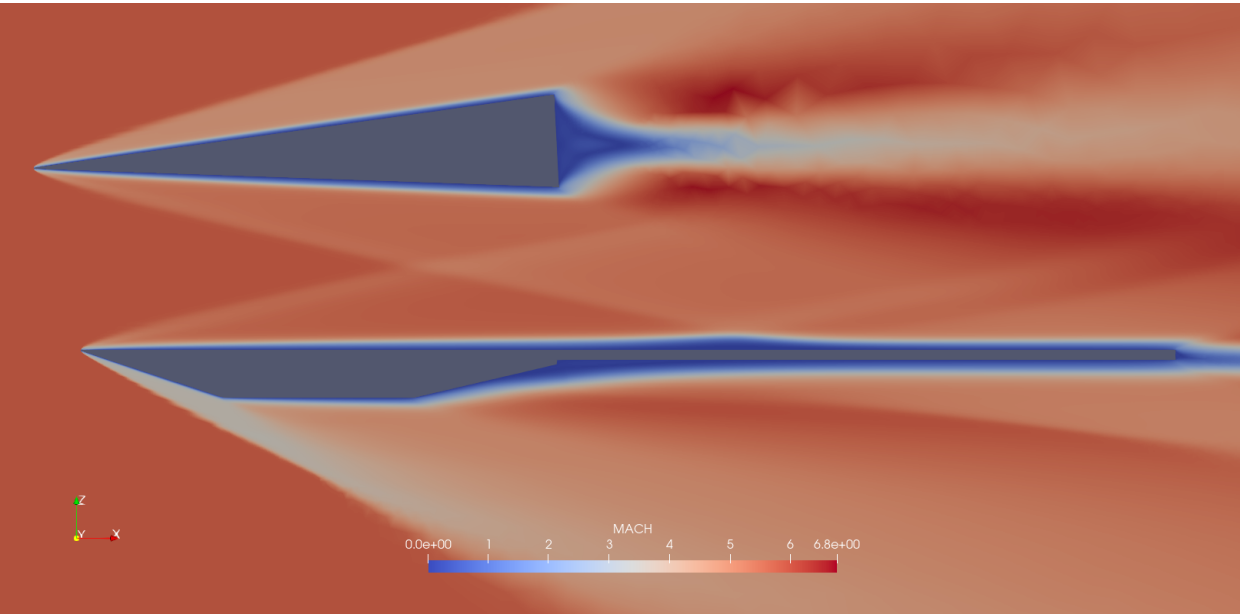
- Fixture and plate included, with fixed boundaries conditions applied to the fixture
- Material properties from the 2019 paper²
 - $E = 52.7 \text{ GPa}$, $\rho = 2670 \text{ kg/m}^3$, $\nu = 0.33$
- Ansys Mechanical used to obtain mode shapes and frequencies
 - 1st mode - 84.723 Hz, 2nd mode - 530.41 Hz
 - From the paper, 1st mode – 83 Hz

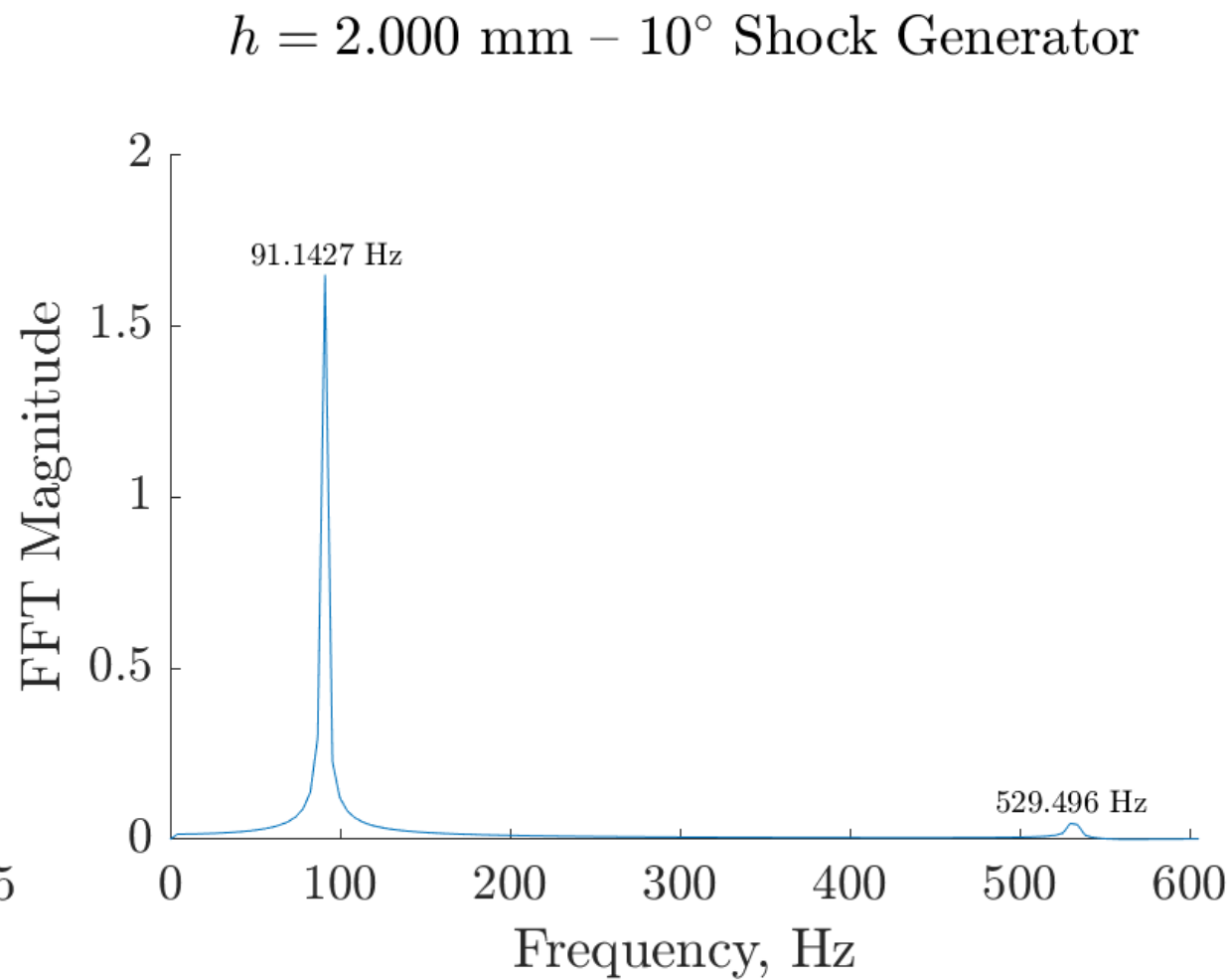
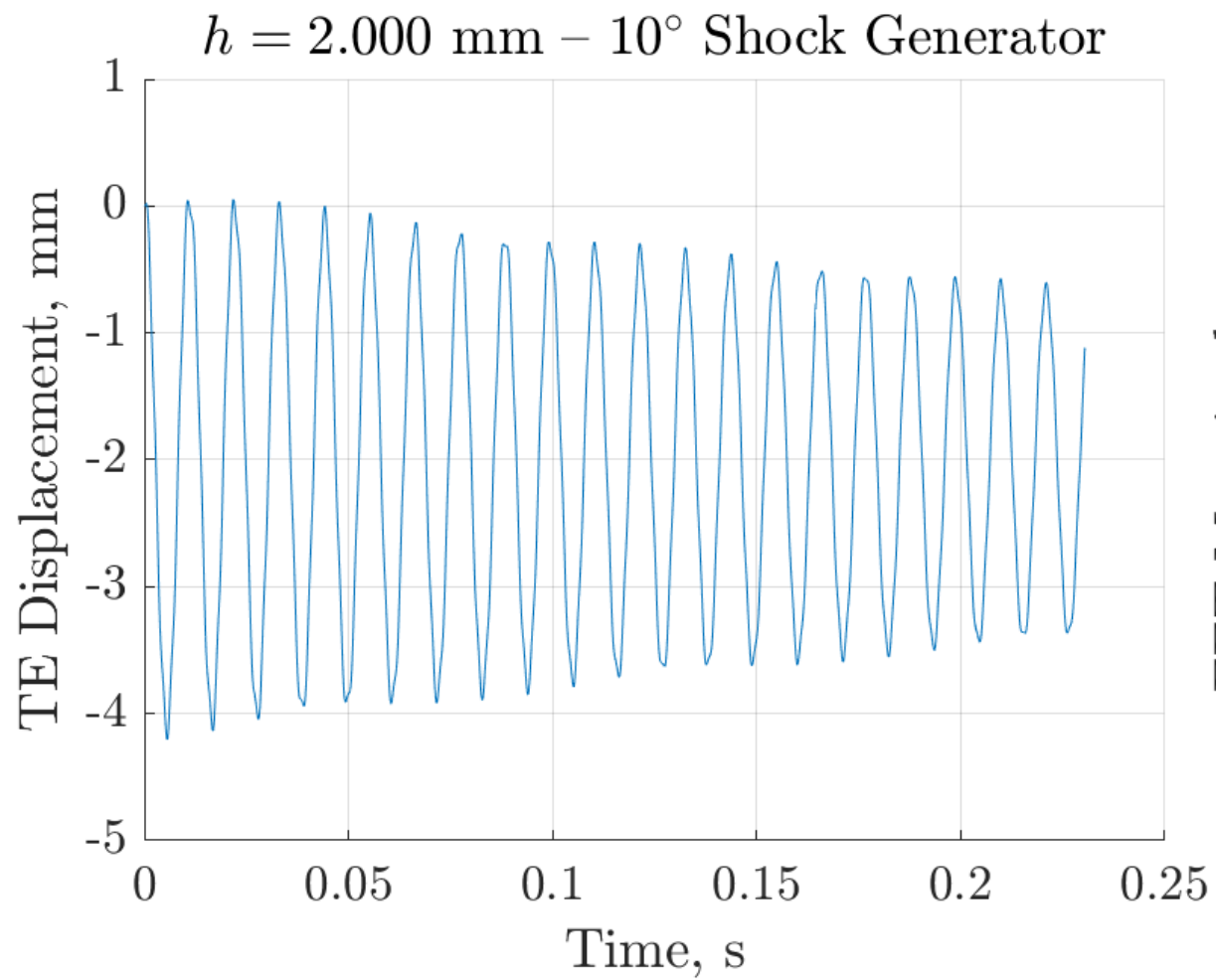
Mach Contours

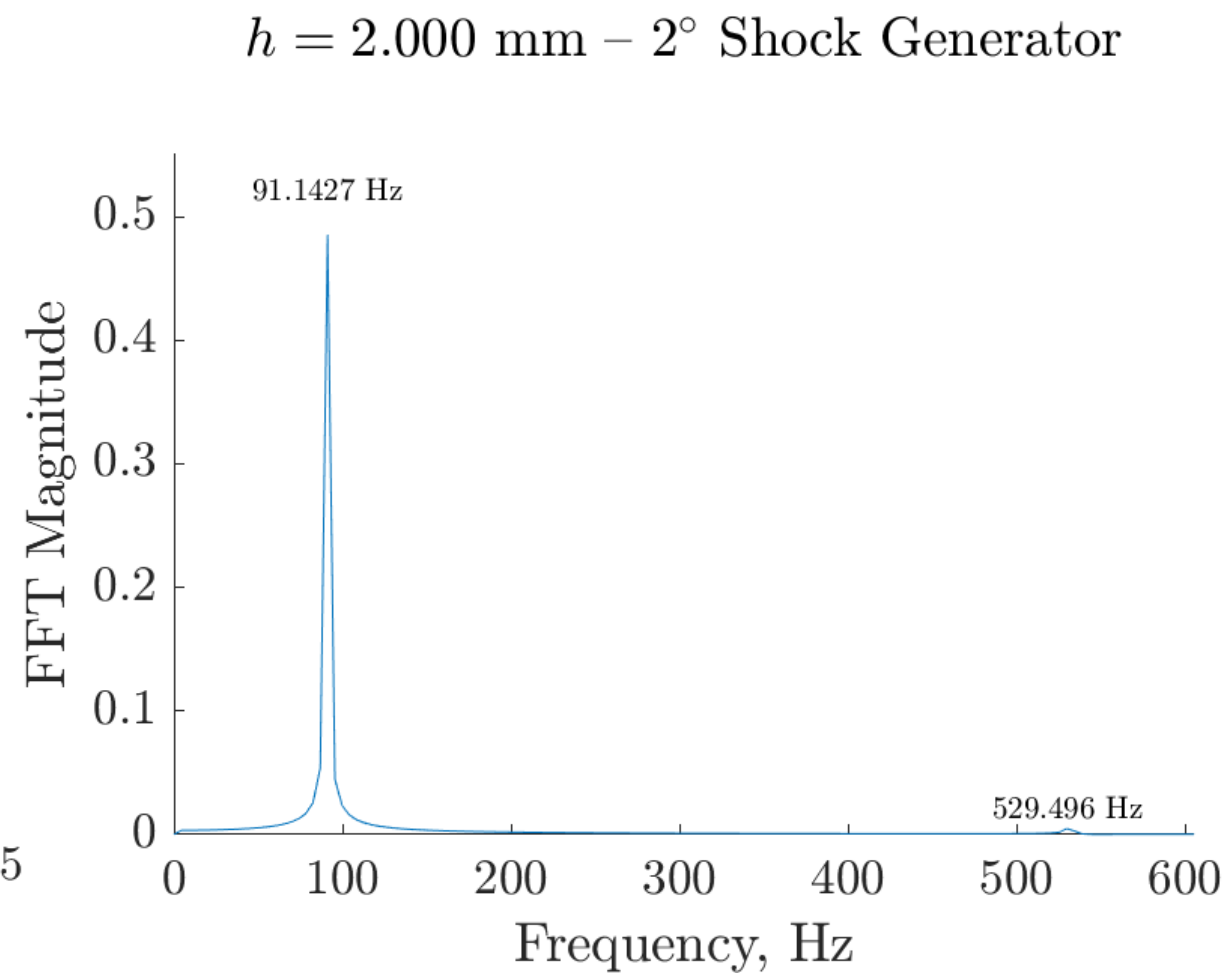
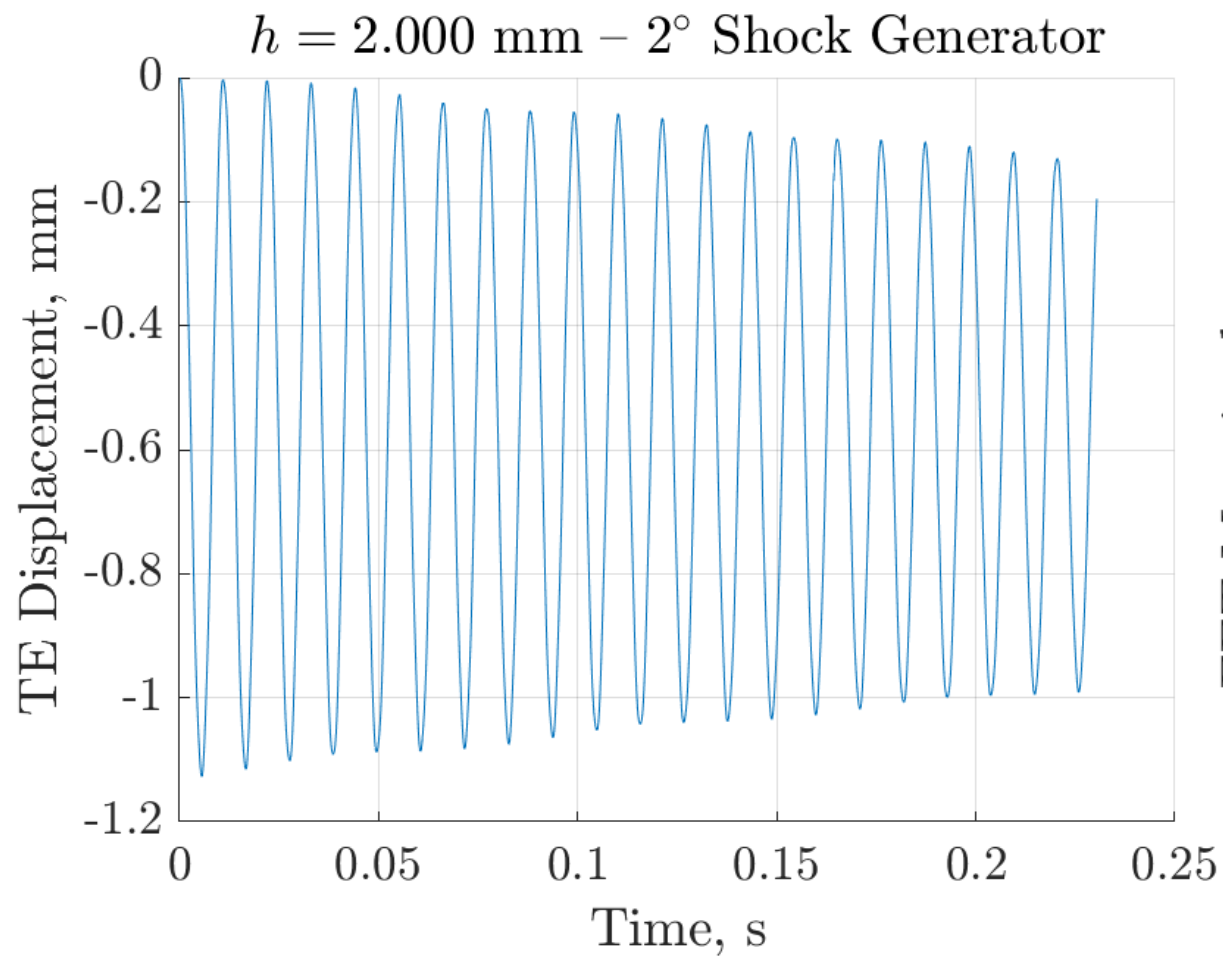
10° Wedge



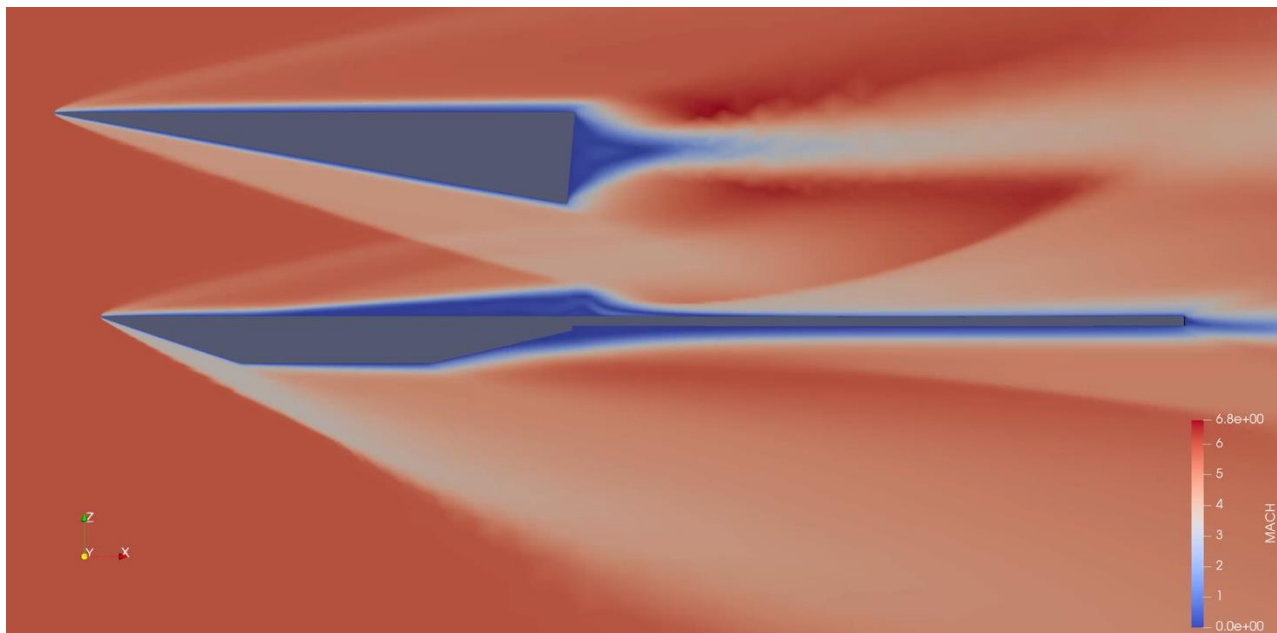
2° Wedge



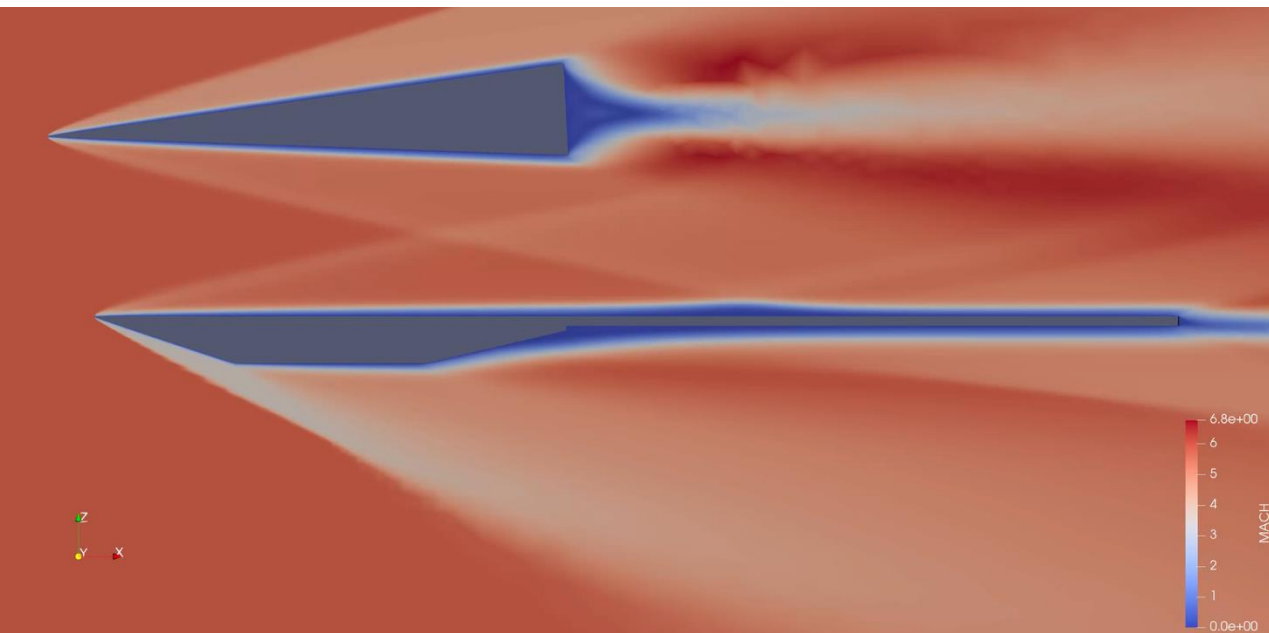




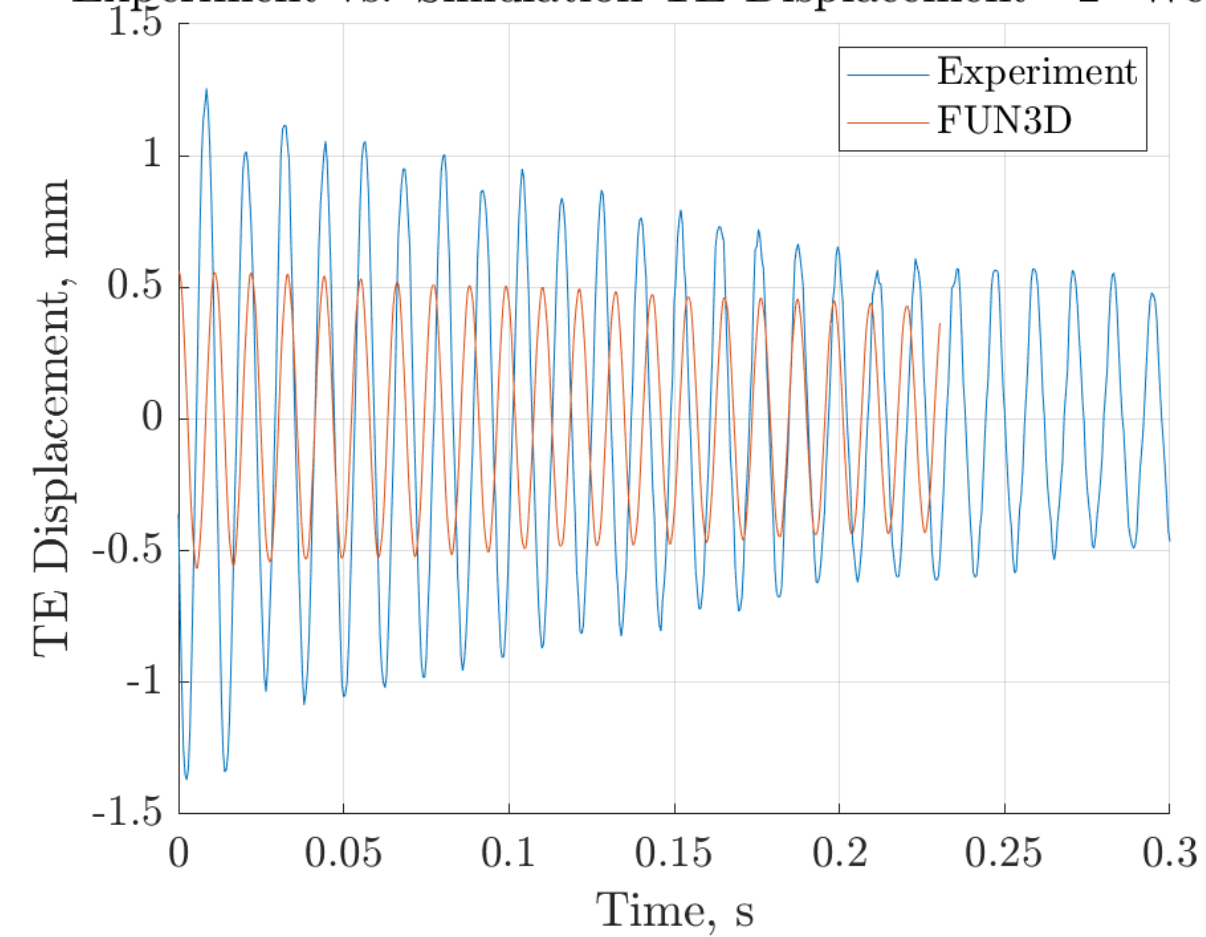
10° Wedge



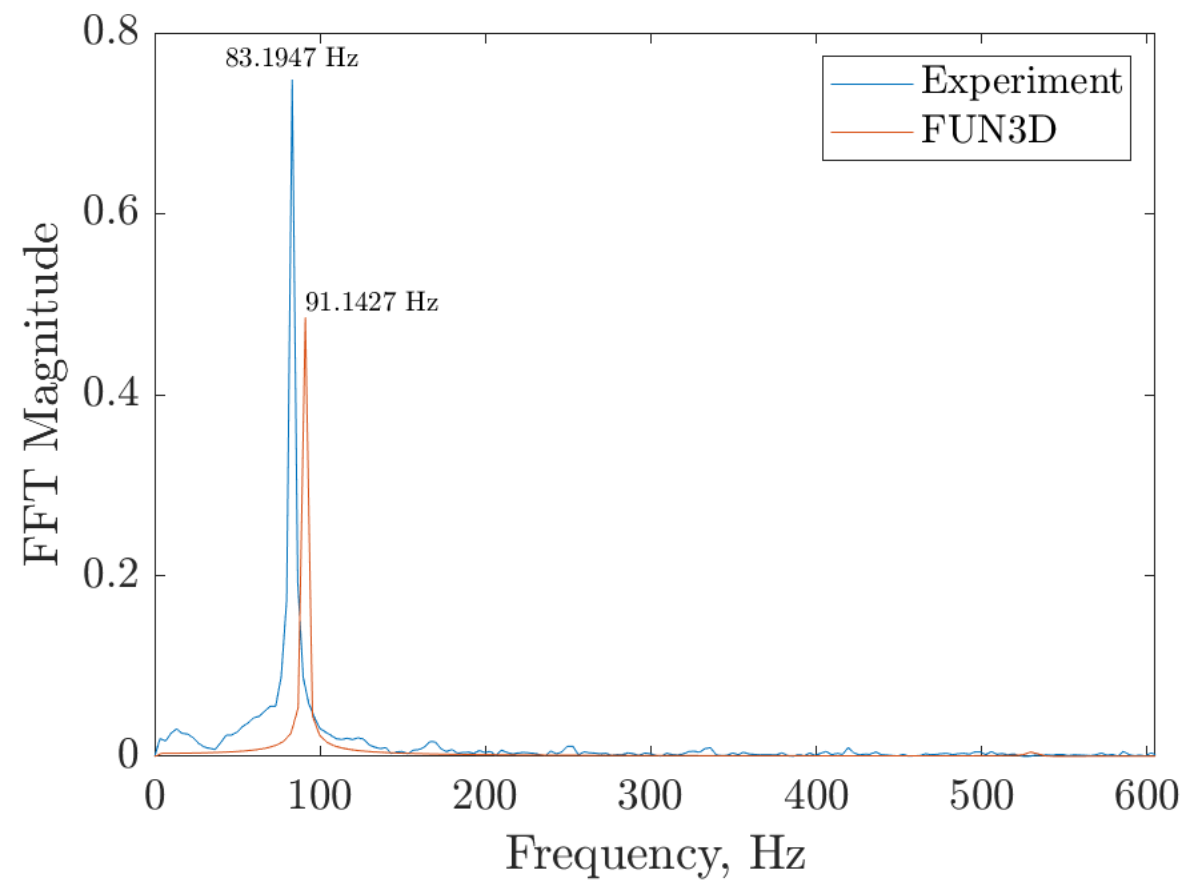
2° Wedge

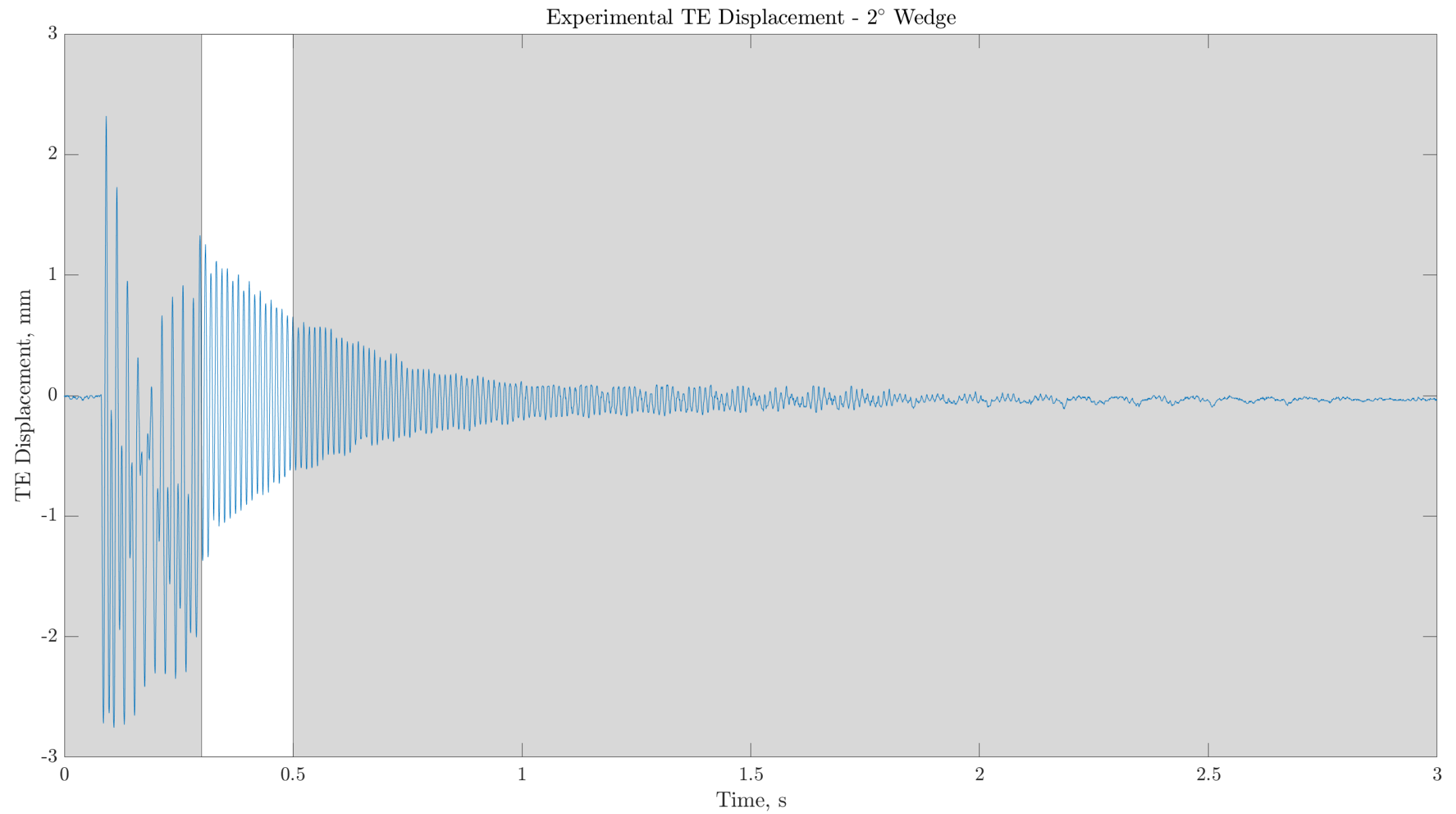


Experiment vs. Simulation TE Displacement - 2° Wedge



$h = 2.000$ mm - 2° Shock Generator





- Predicting flutter boundary through an analytical parameter
 - Strong sensitivity to aerodynamic to structural compliance ratio, shown by McHugh et al.³

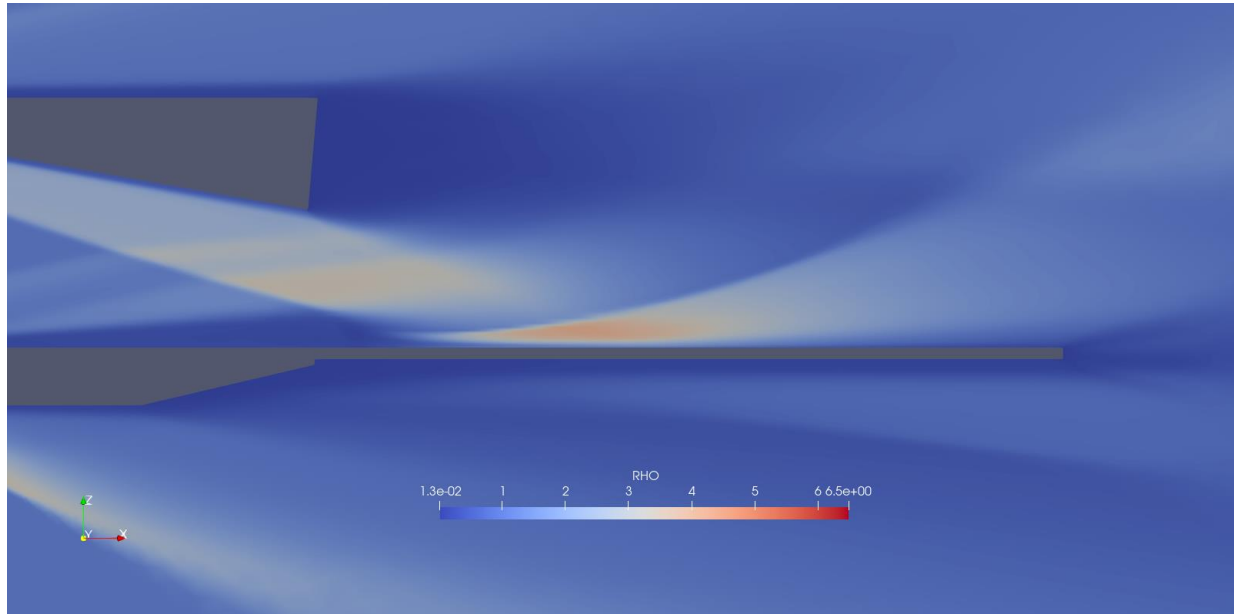
$$\Lambda = \frac{\rho U^2 L^3}{MD}$$

$$D = \frac{Eh^3}{12(1 - \nu^2)}$$

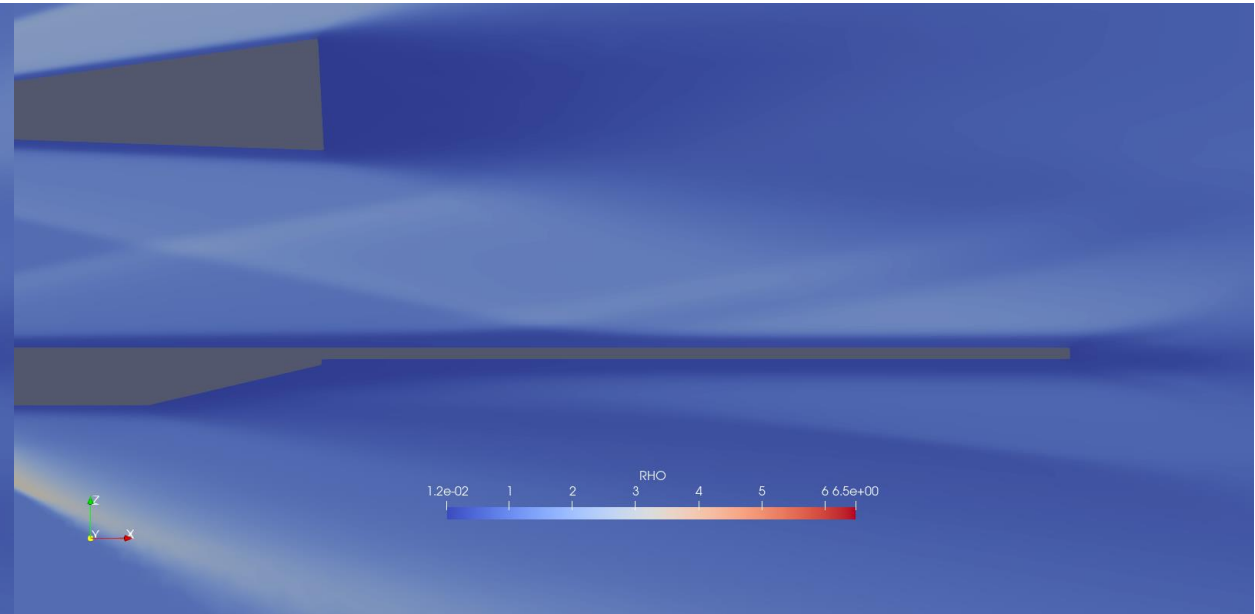
- Depends on flow parameters
 - Free-stream vs. local parameters (post-shock impingement)

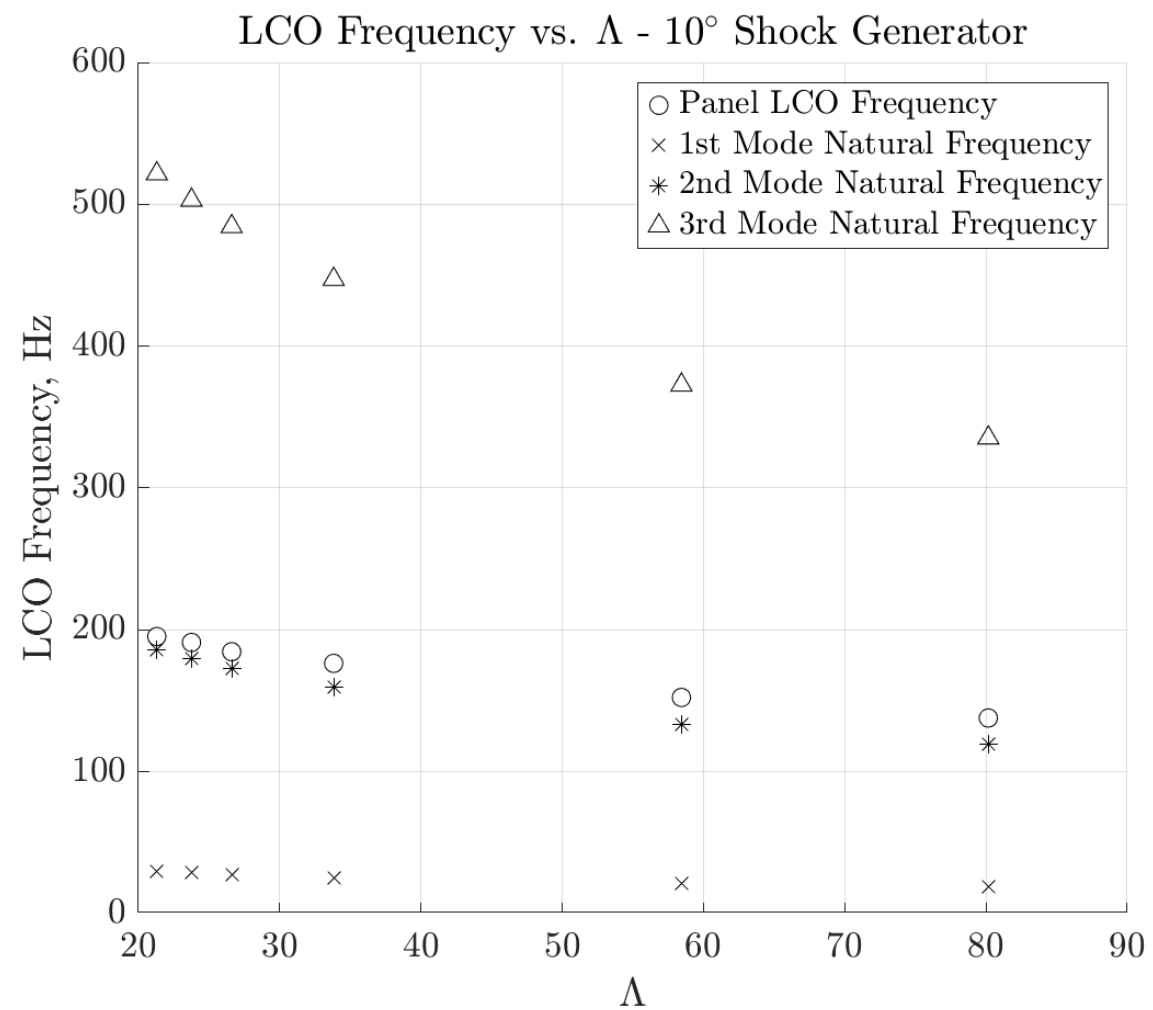
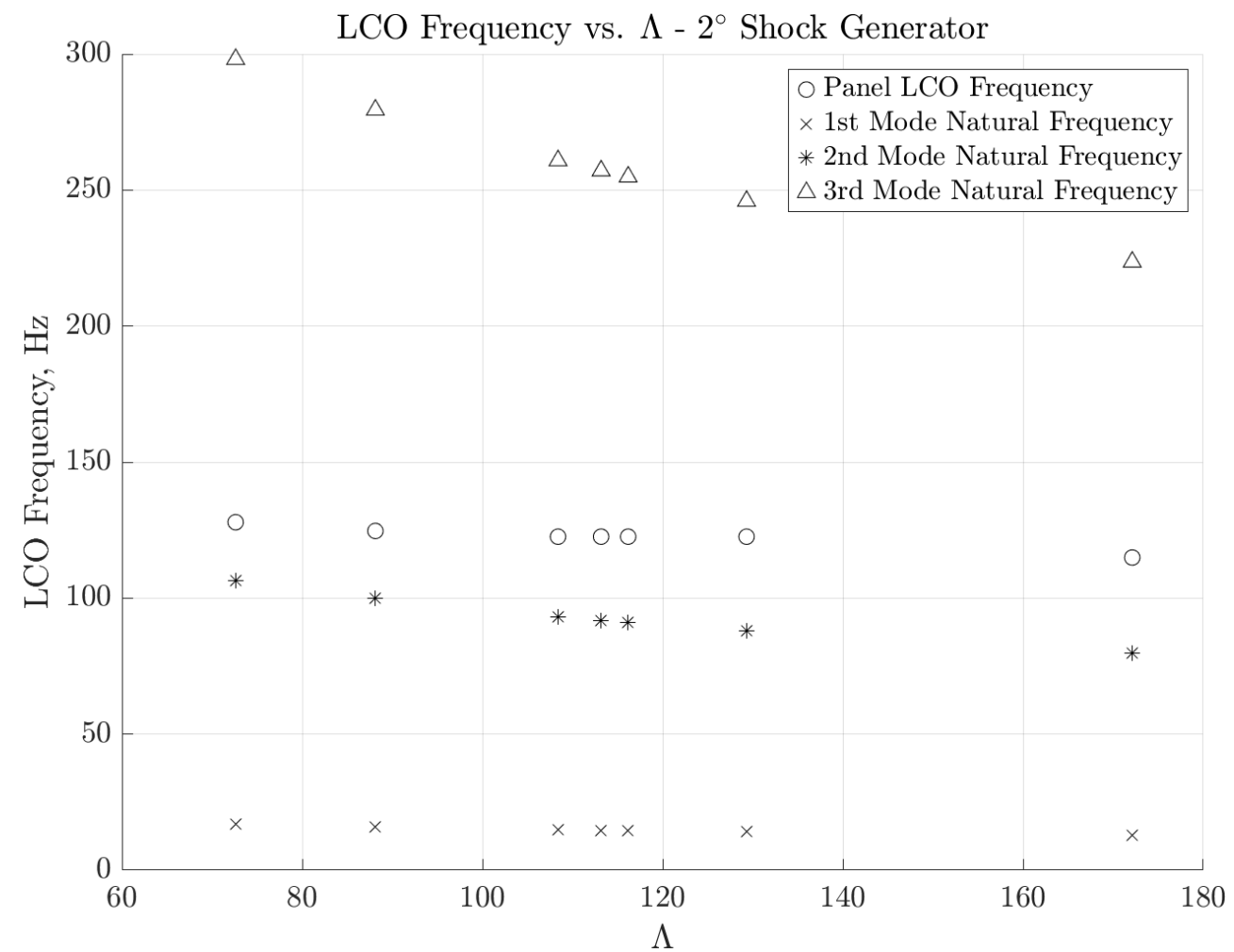
Density Contours

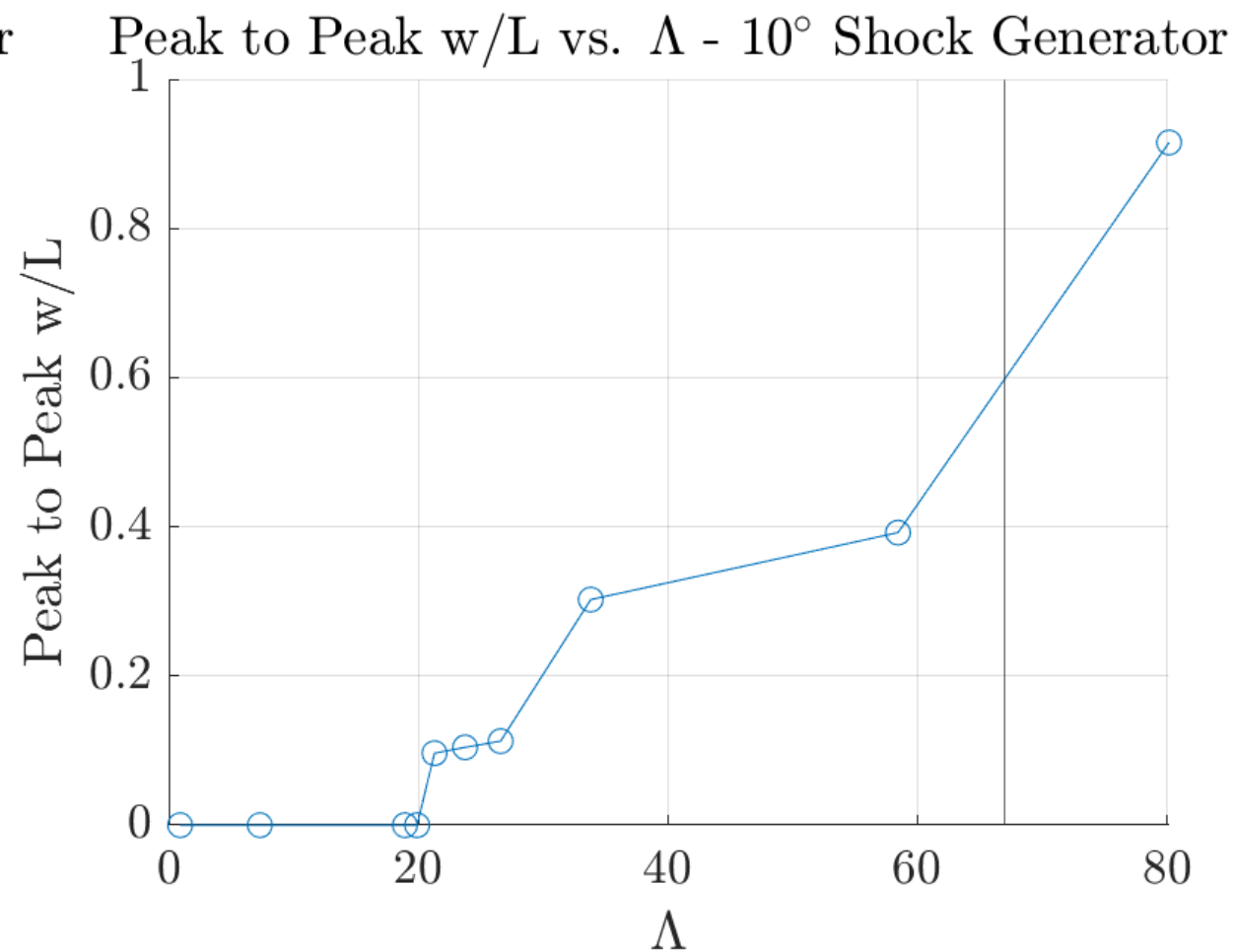
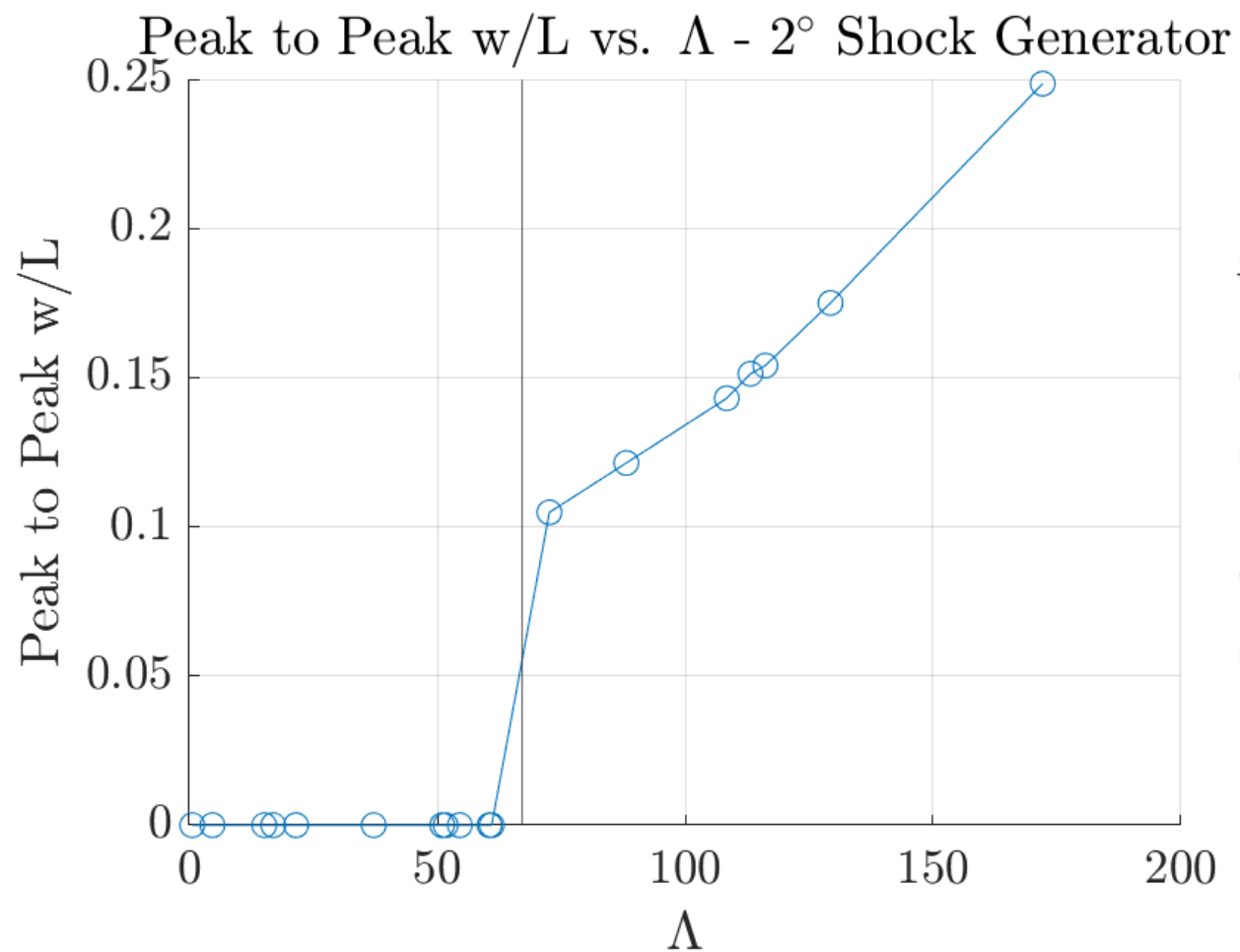
10° Wedge

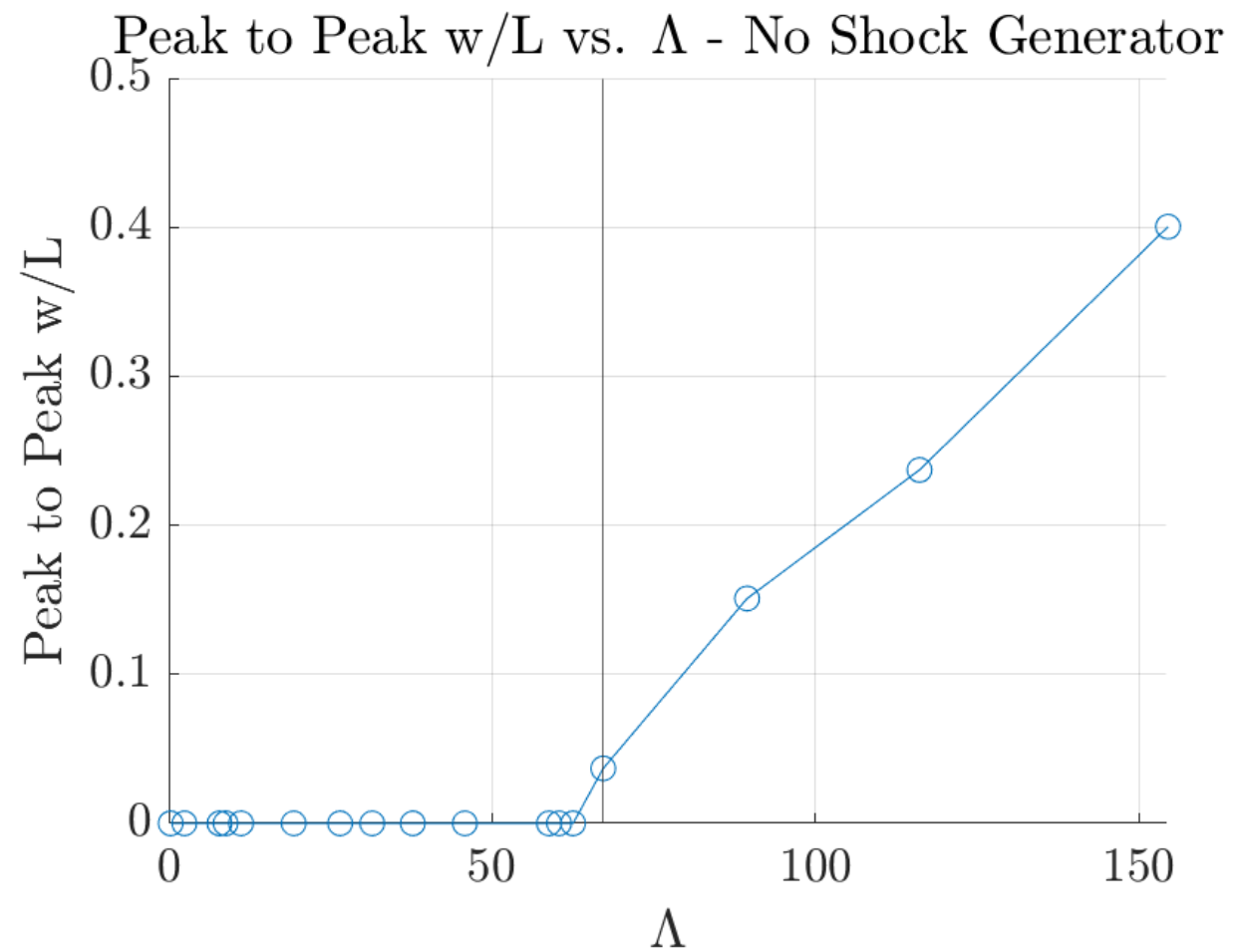
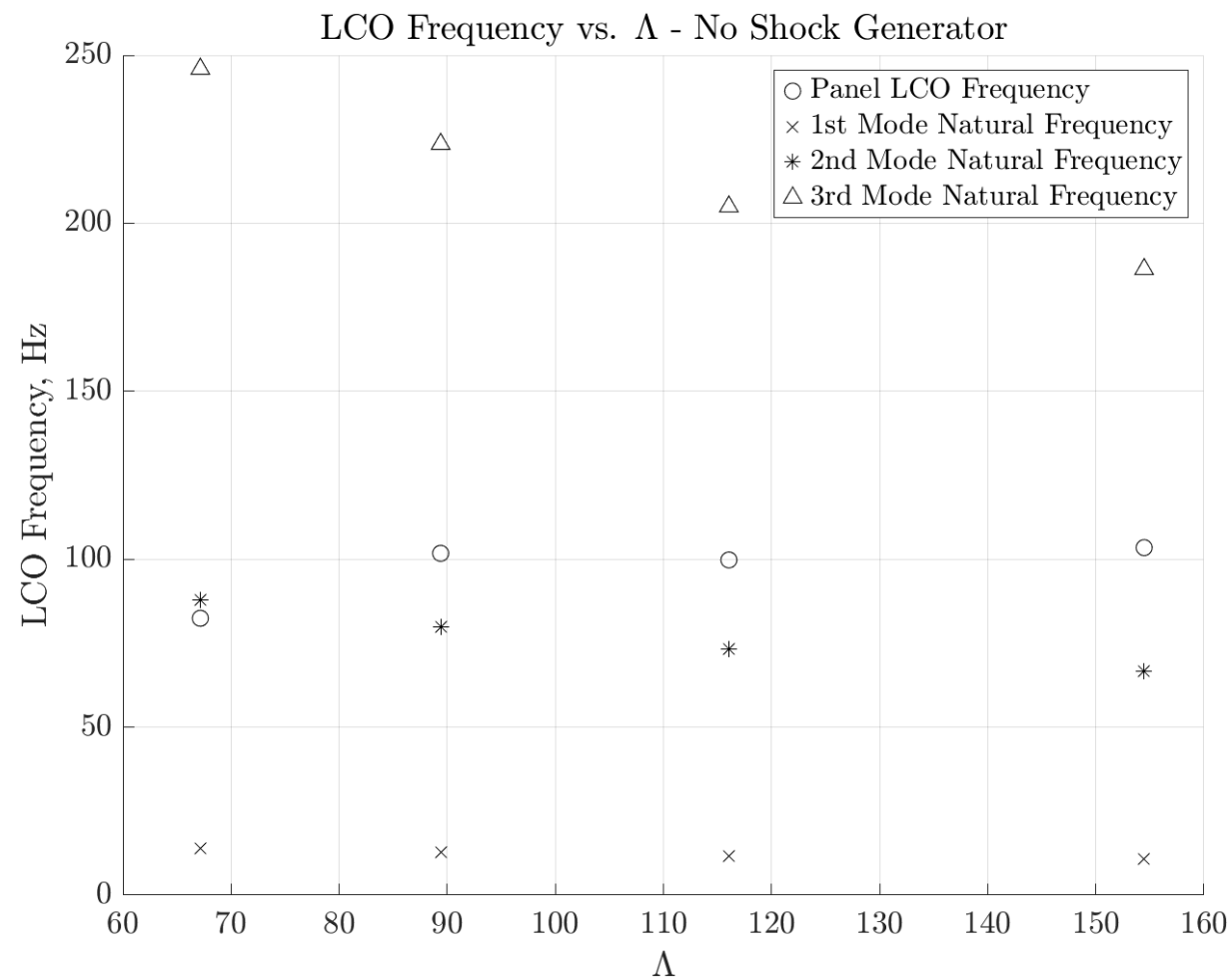


2° Wedge





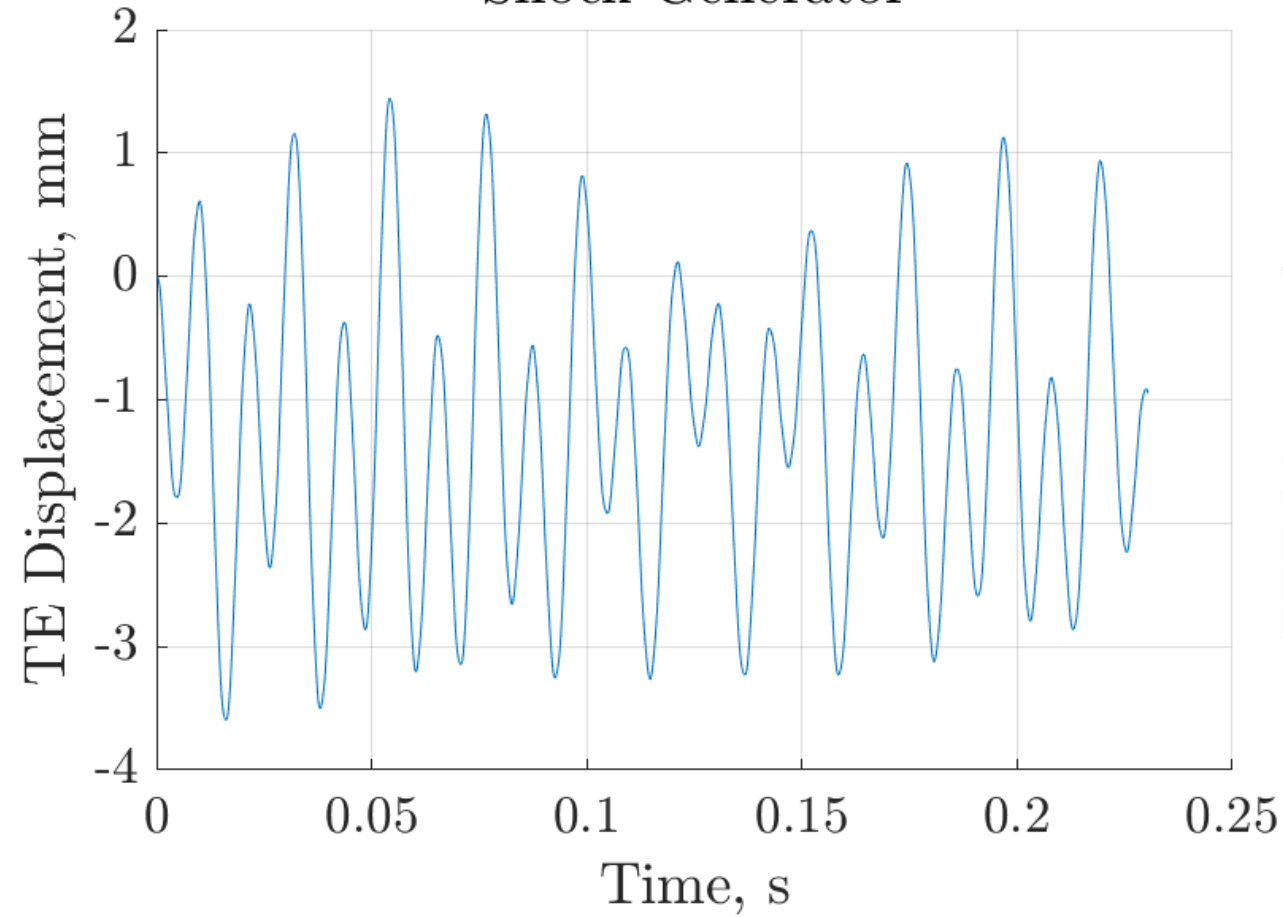




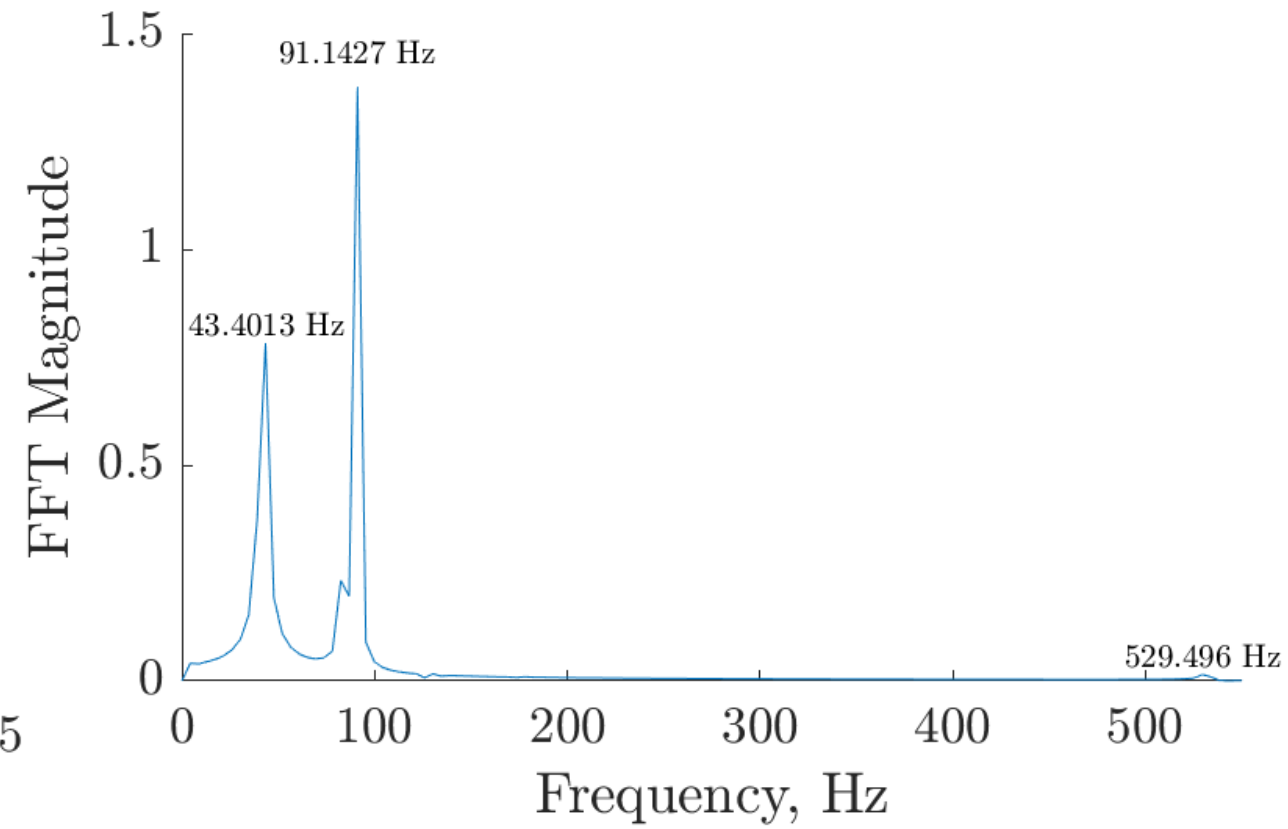
Oscillating Wedge Case

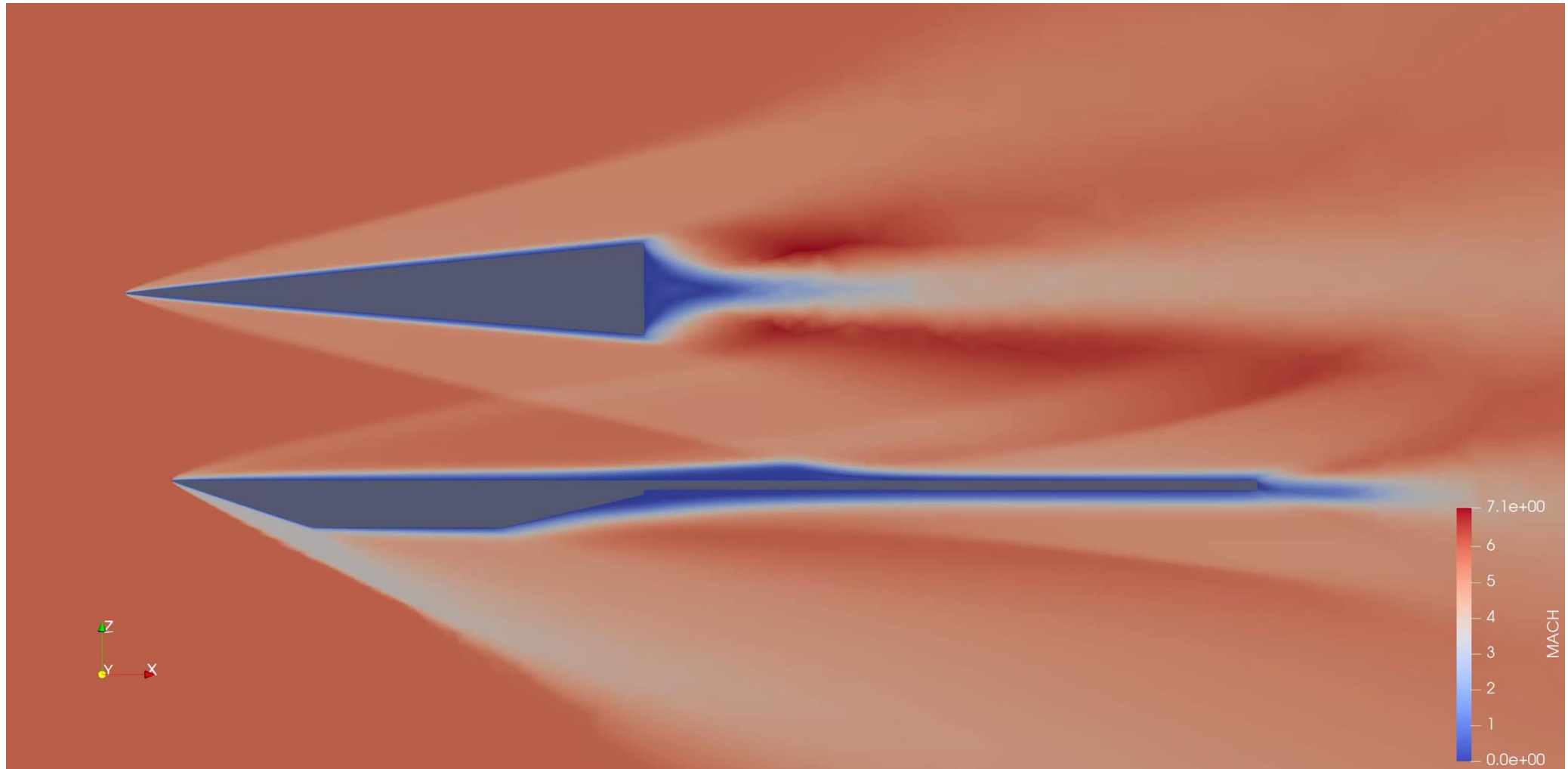
- An optional test case from AePW3
 - Wedge geometry was the same as in the data package
 - The wedge pitches at a frequency of 42 Hz⁴
 - Starts at 5° and oscillates $\pm 3^\circ$

$h = 2.000 \text{ mm} - 5^\circ$ Oscillating
Shock Generator



$h = 2.000 \text{ mm} - 5^\circ$ Oscillating
Shock Generator





Conclusions

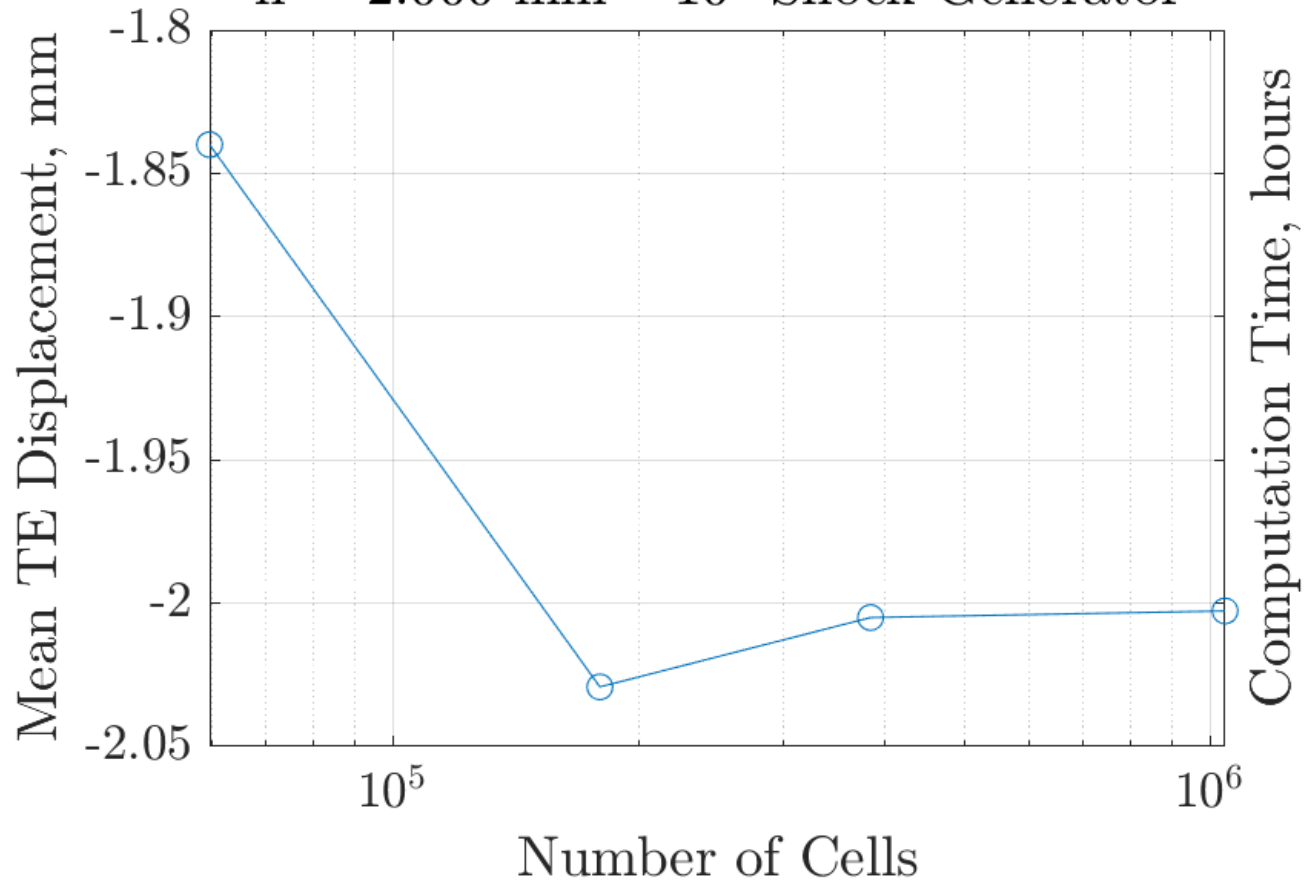
- FUN3D Simulations of the HyMAX experiment have been performed
 - For the experimental plate of a certain thickness studied by Currao et al., our results are consistent with that of others, i.e. a stable decaying oscillation of the plate with a frequency near the first natural frequency.
 - Simulations for thinner plates show flutter and limit cycle oscillations (LCO) with and without a shock generator.
 - Without a shock generator, the flutter and LCO results are similar to those found by Kevin McHugh using piston theory aerodynamics.
 - With a shock generator, thicker plates are needed to suppress flutter and LCO. The thickness needed to suppress flutter/LCO increases with increases in shock generator wedge angle.

Future work

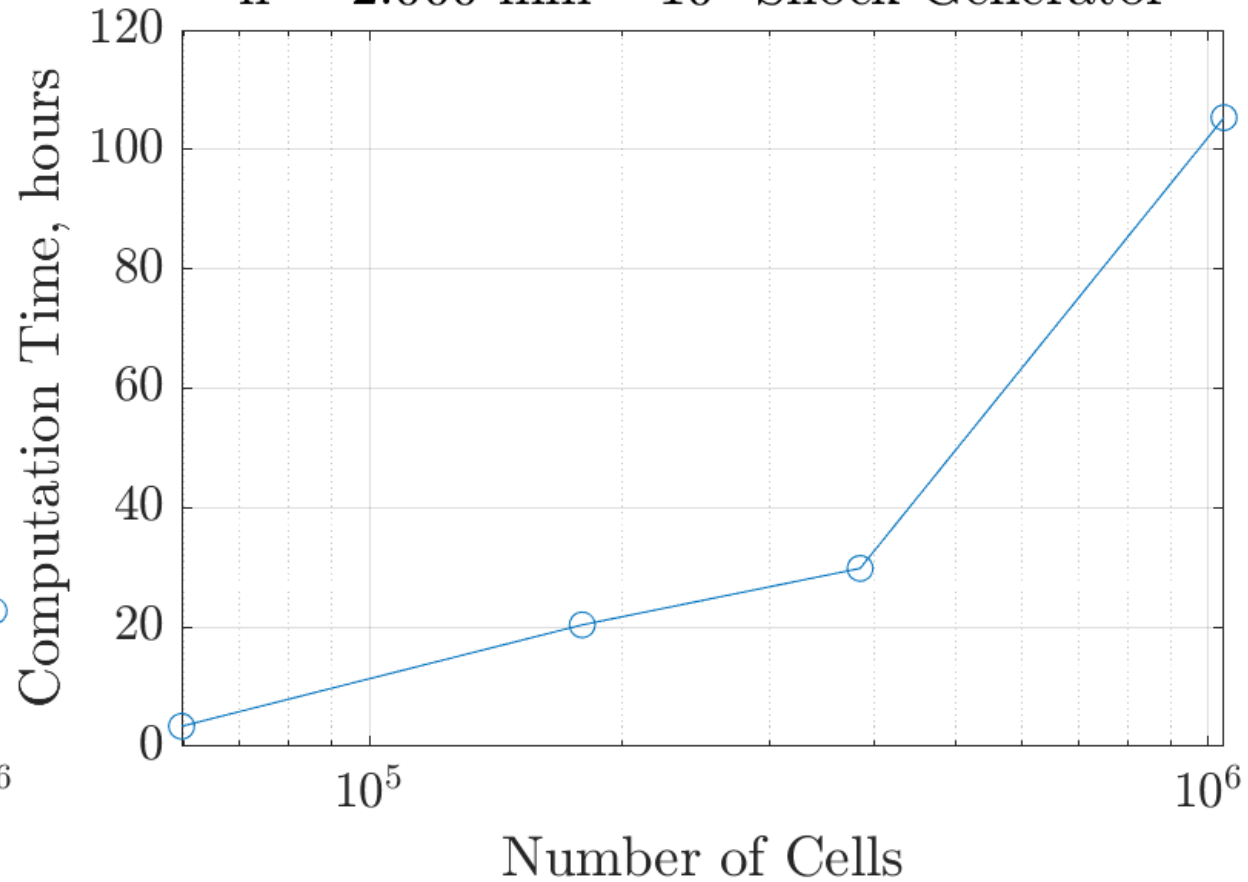
- Utilize the FUNtoFEM framework to couple with a FEM solver and compare with current internal modal solver results
 - Use additional modes with modal solver
- Replicate the tunnel start-up through different imposed initial conditions
- Introduce thermal effects to the panel
- Extrude the domain in the spanwise direction and run the simulation as 3D

Backup Slides

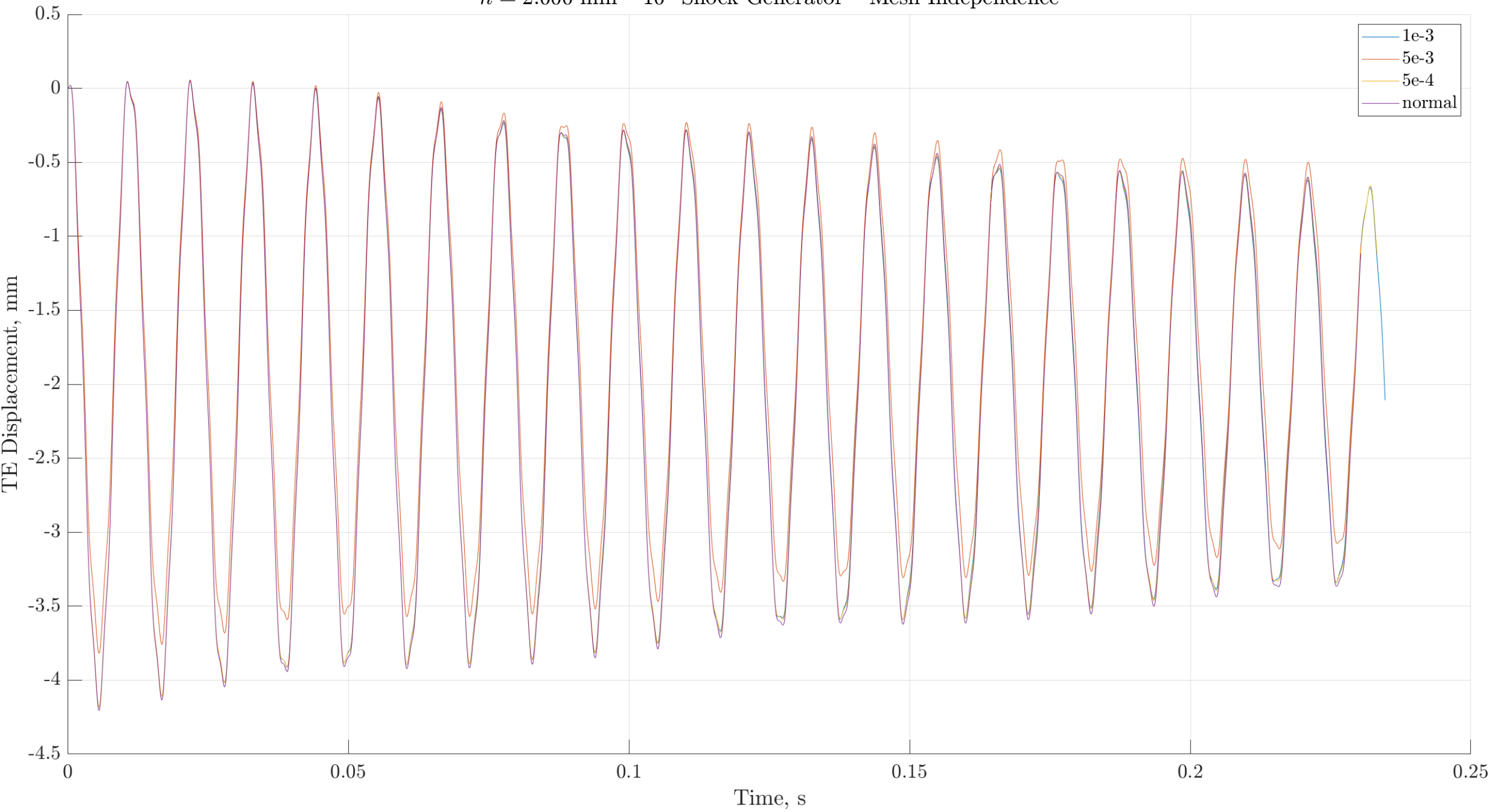
Grid Convergence Study
 $h = 2.000 \text{ mm} - 10^\circ$ Shock Generator

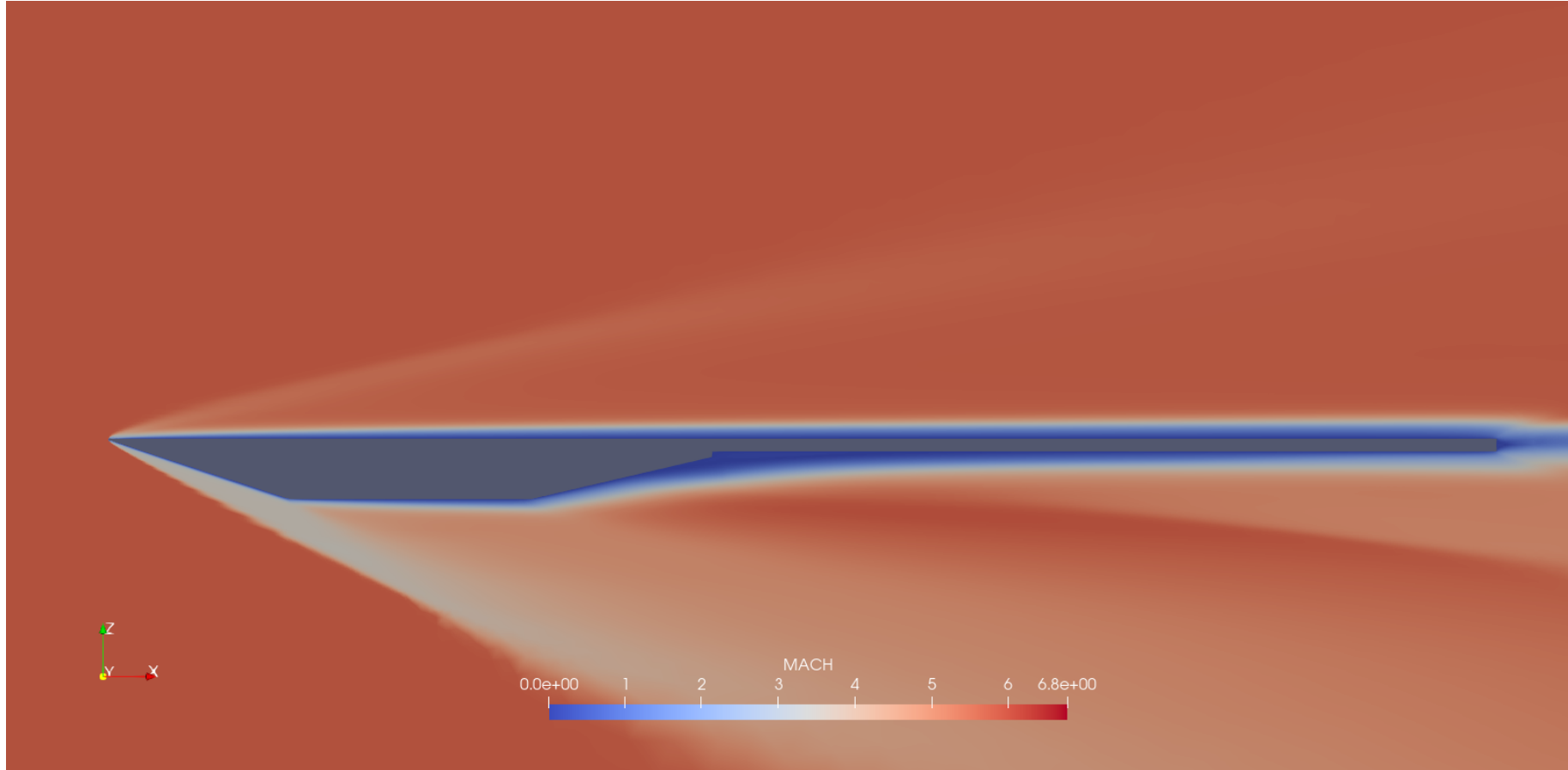


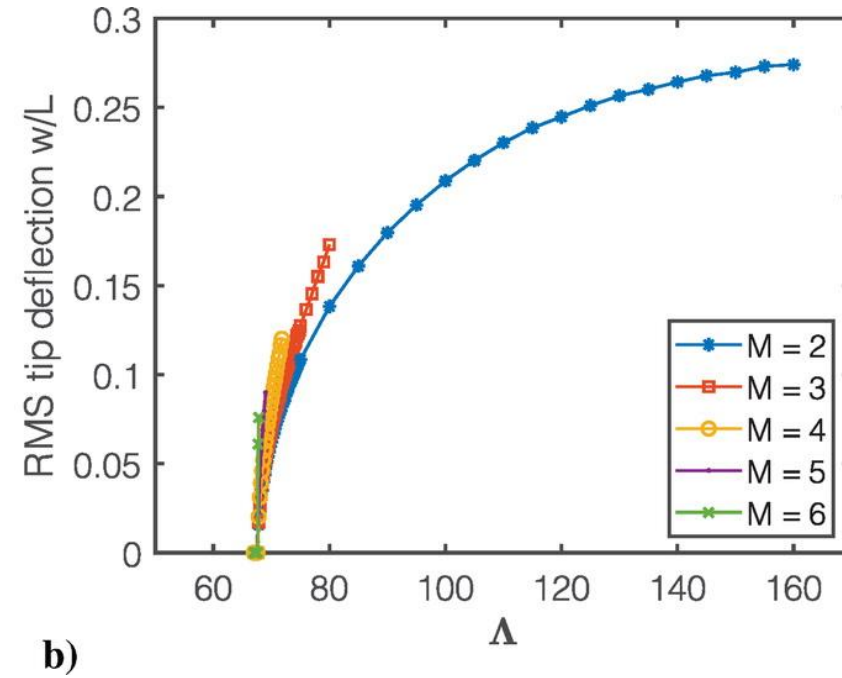
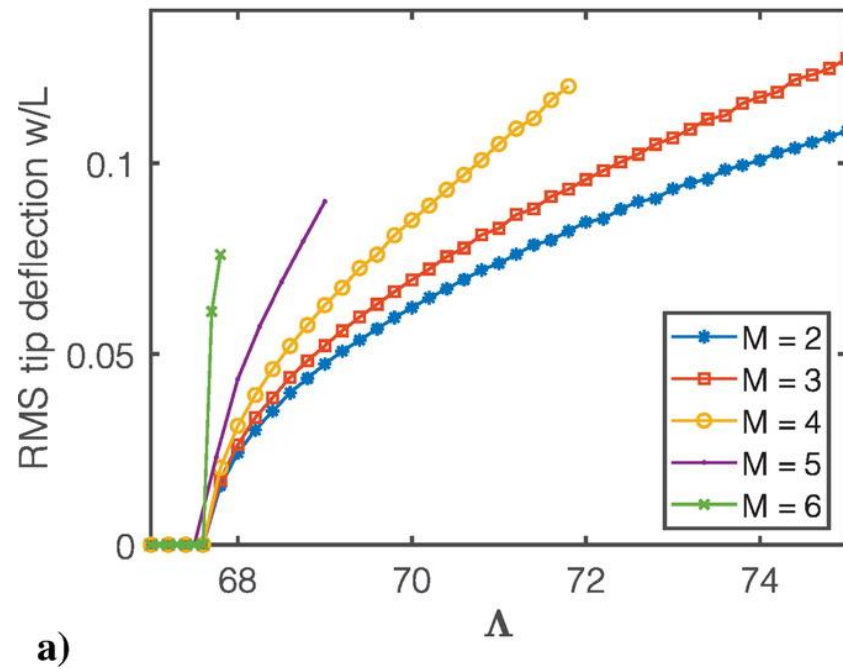
Grid Computation Time
 $h = 2.000 \text{ mm} - 10^\circ$ Shock Generator

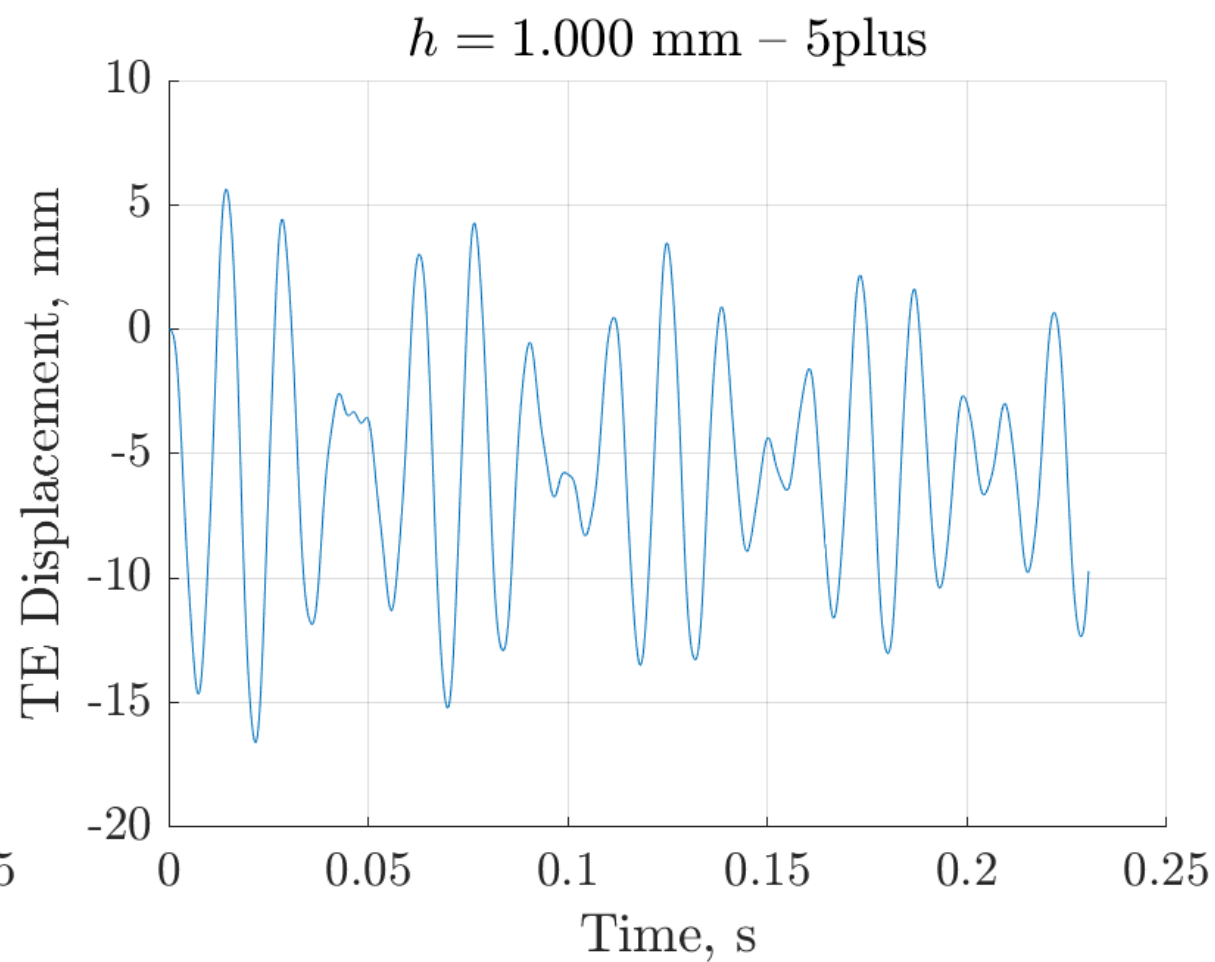
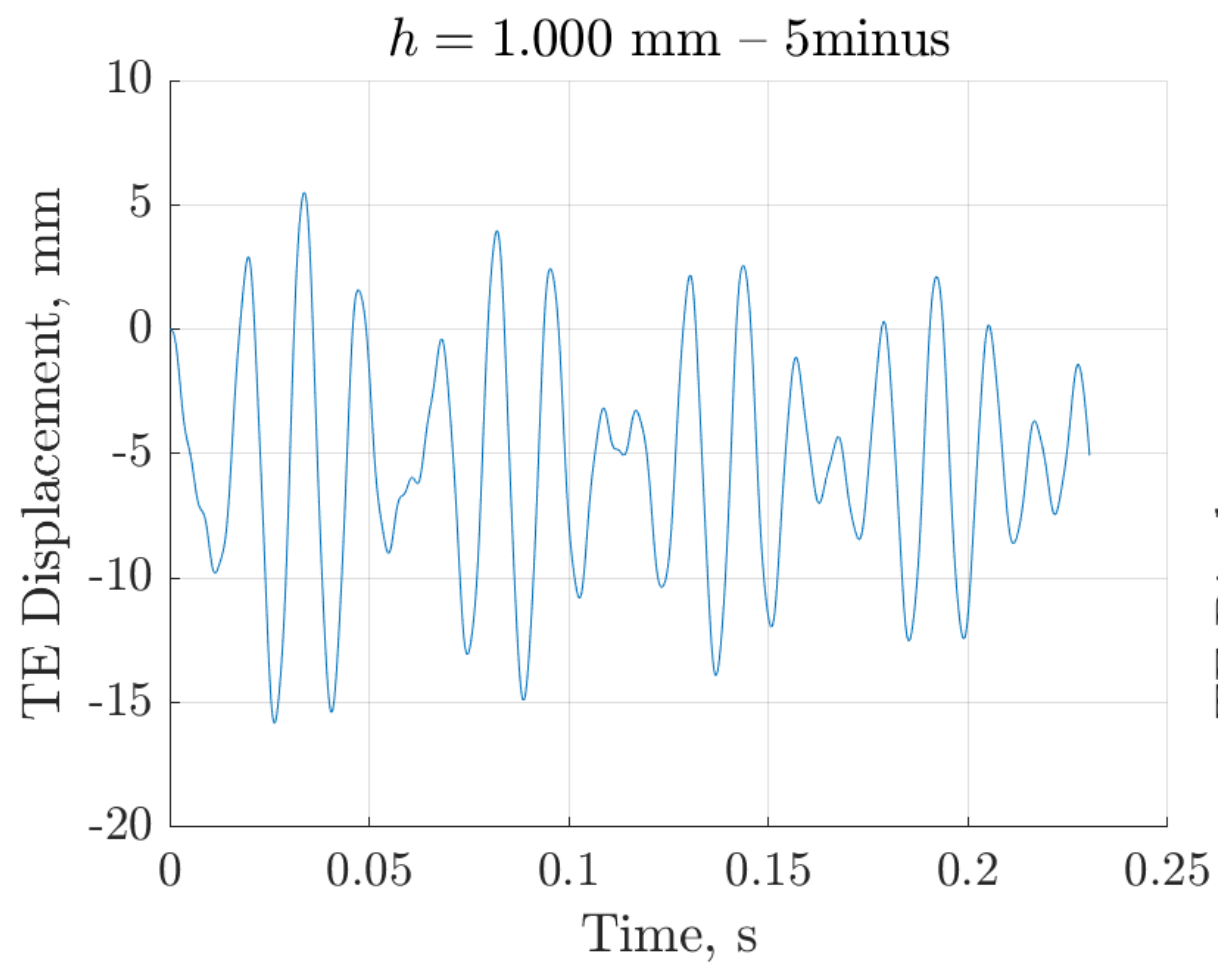


$h = 2.000 \text{ mm} - 10^\circ$ Shock Generator – Mesh Independence

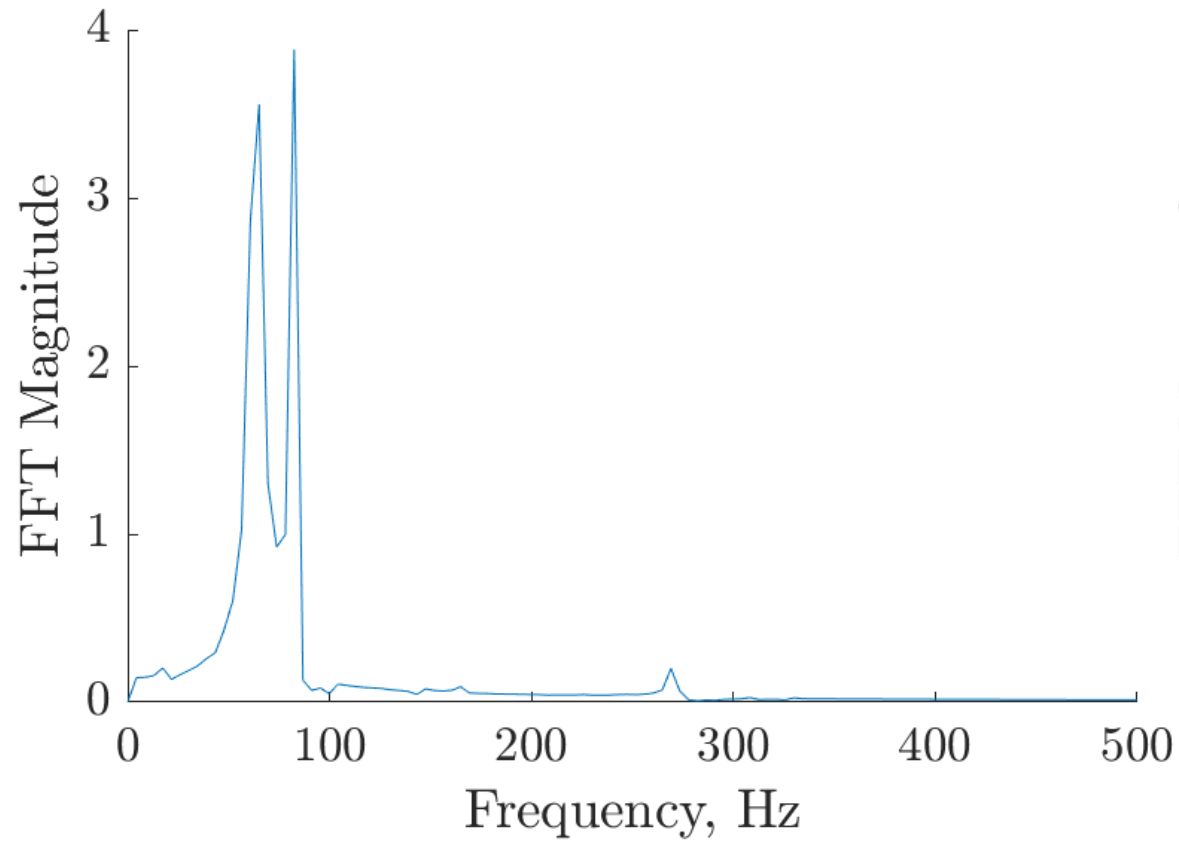




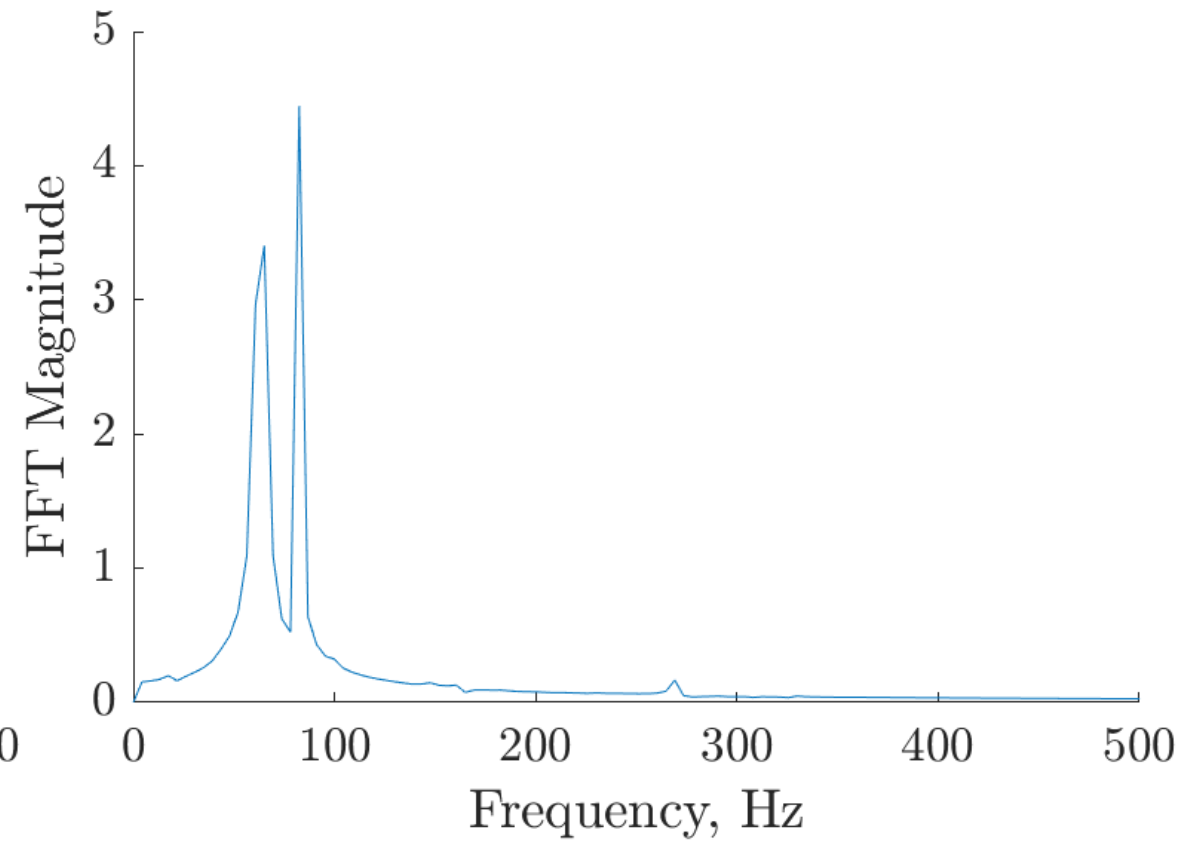




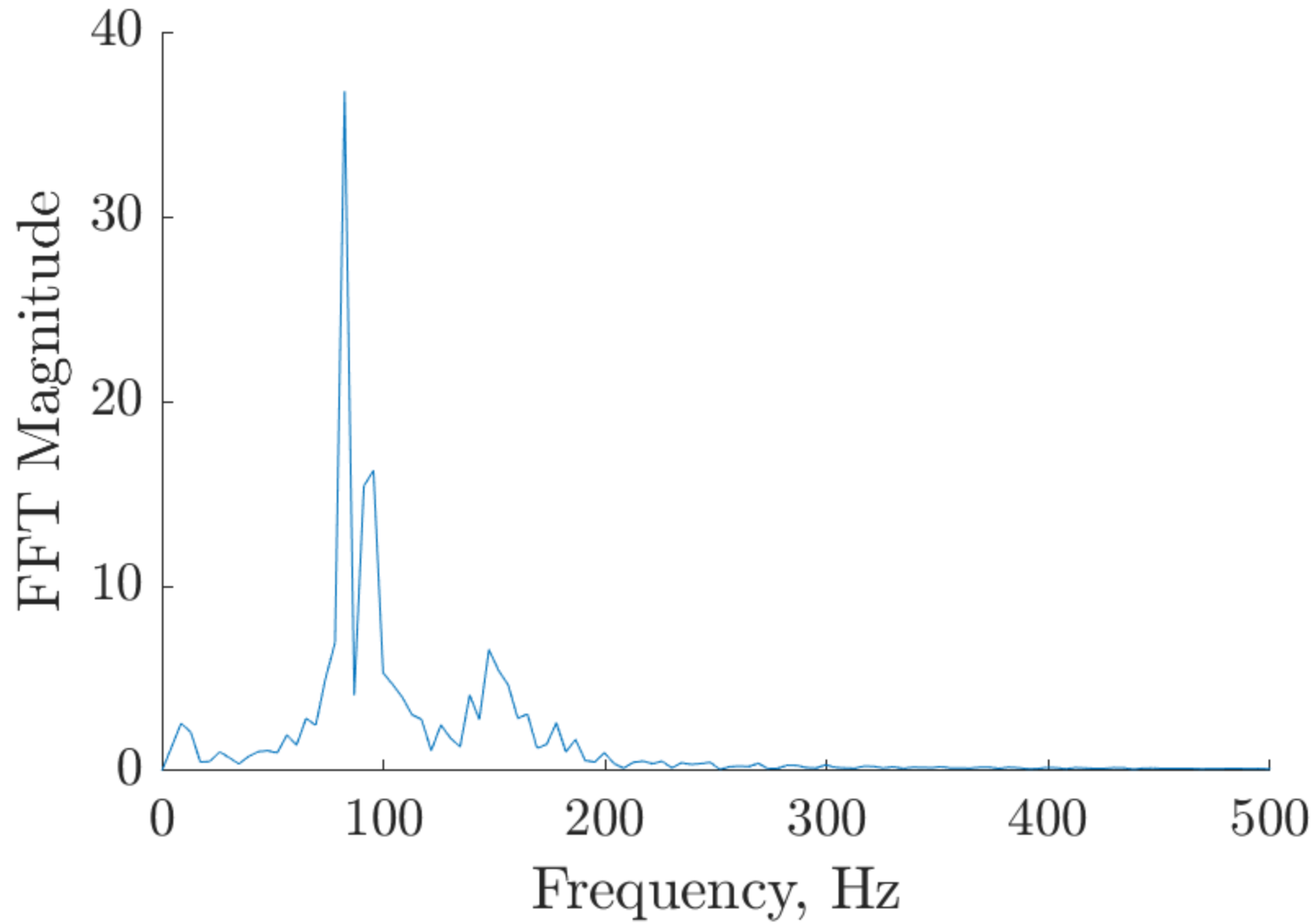
$h = 1.000 \text{ mm} - 5\text{minus}$

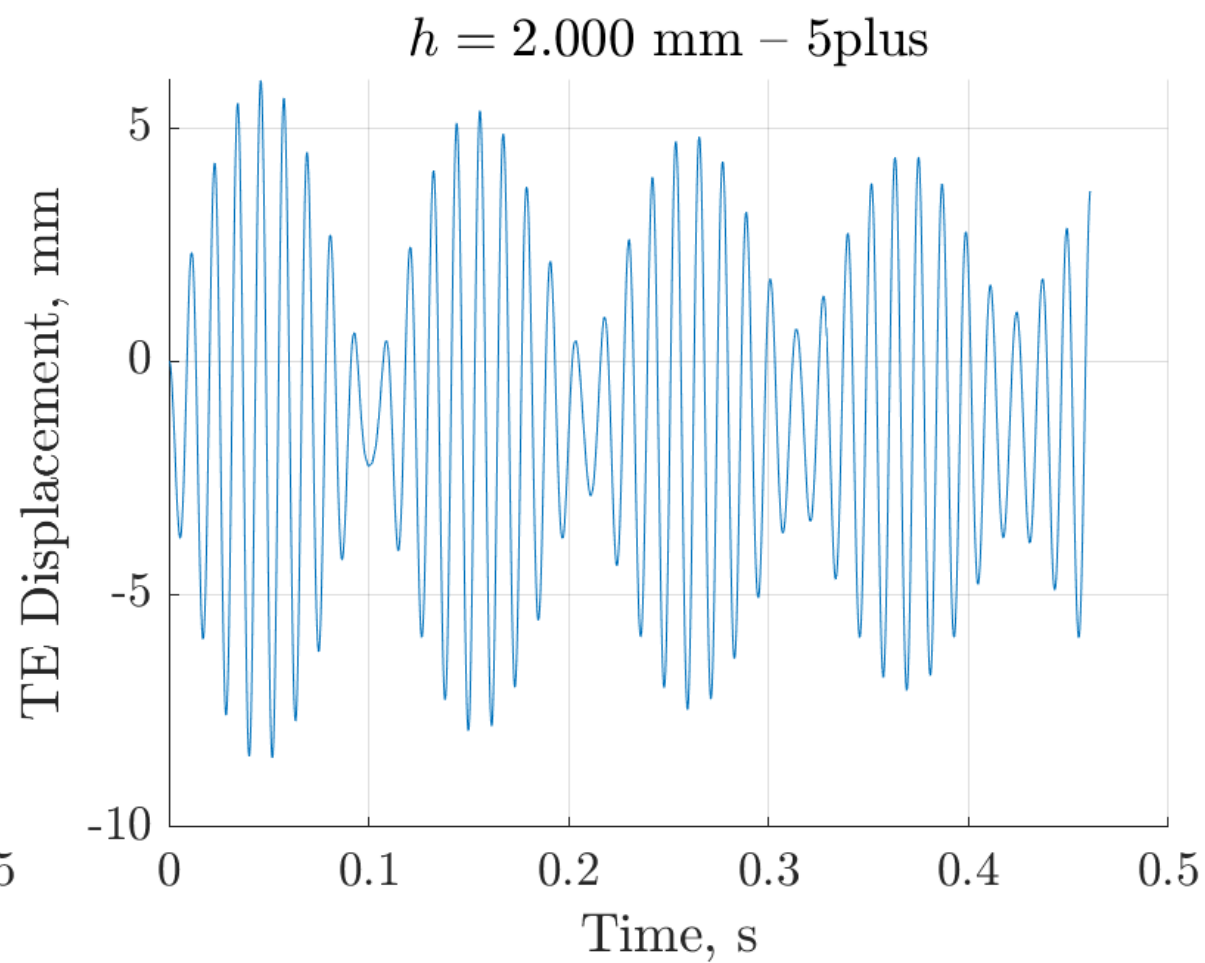
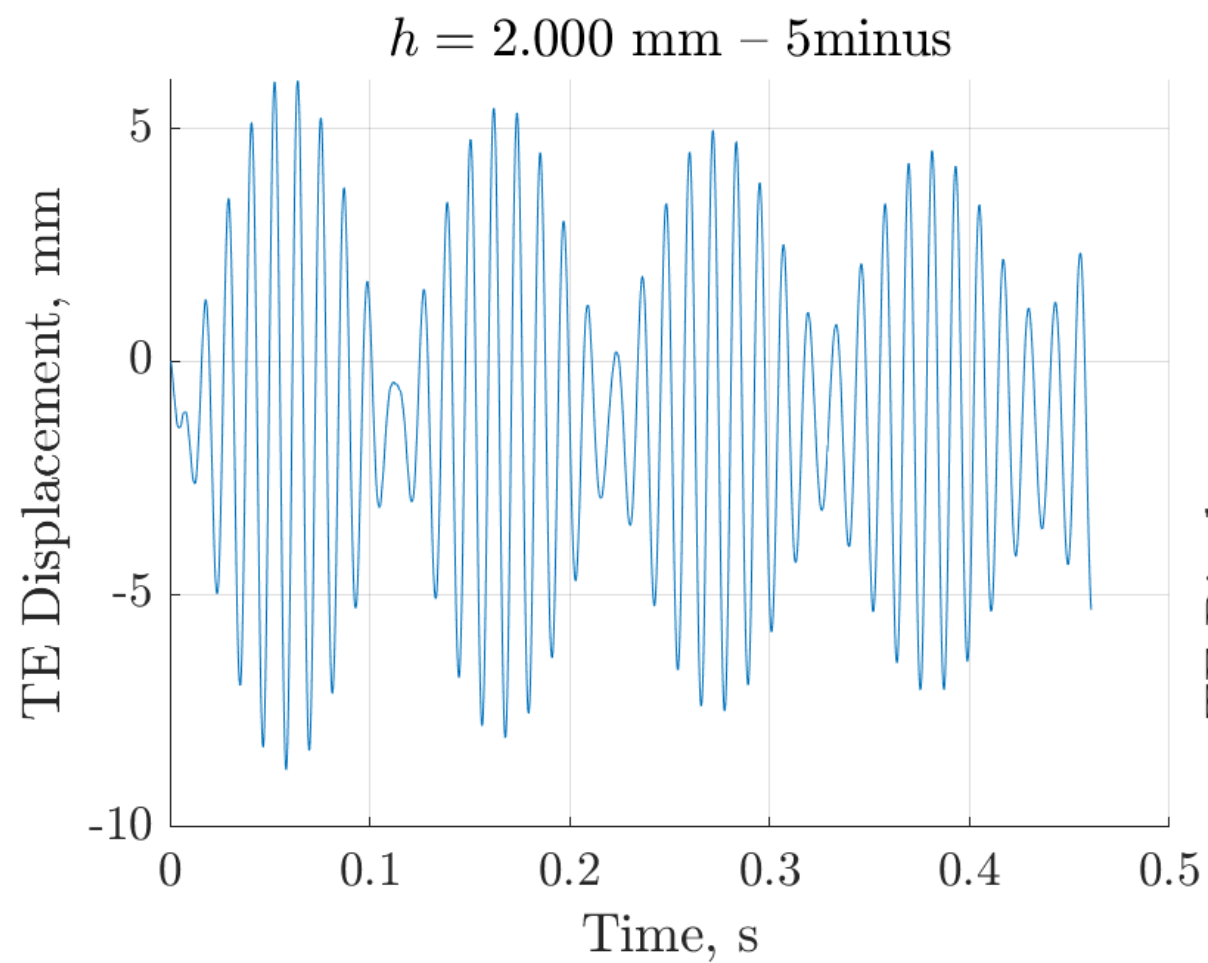


$h = 1.000 \text{ mm} - 5\text{plus}$

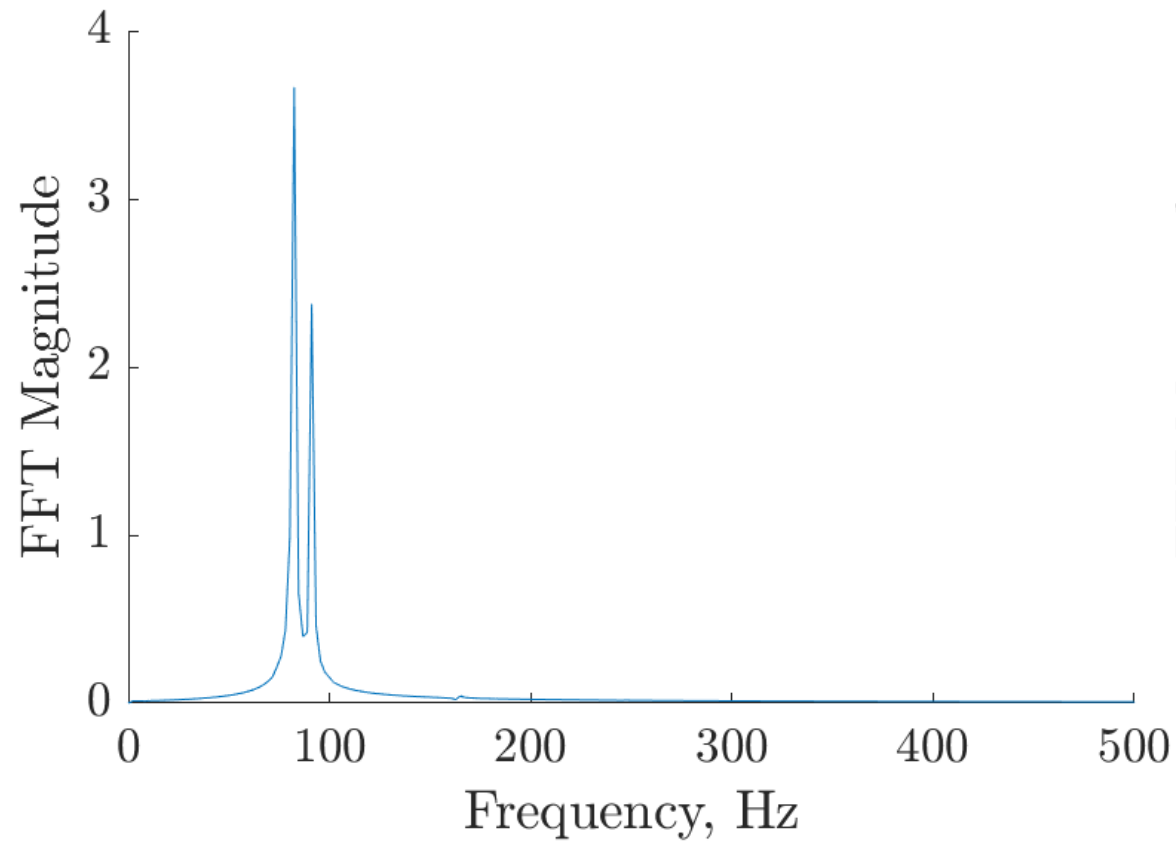


$$h = 0.500 \text{ mm} - 5\text{minus}$$





$h = 2.000 \text{ mm} - 5\text{minus}$



$h = 2.000 \text{ mm} - 5\text{plus}$

